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Review

Advances in Traffic Accident Prediction: A Survey of Novel Approaches

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Abstract

Traffic accidents significantly impact societies and economies. The risk of collision is highest in urban areas, leading to devastating loss of life and escalating socioeconomic costs. In this context, numerous studies have focused on accurately predicting accident risk, severity, and duration using various methodologies. This paper presents an overview of traditional statistical models for accident prediction and a comprehensive systematic review of the literature on statistical modeling, machine learning (ML), and deep learning (DL) techniques employed in this field. Different methodologies and techniques are compared by categorizing studies that adopt similar approaches and analyzing them comparatively. Furthermore, a distinction is made between temporal and spatiotemporal models to describe how these approaches influence the accuracy of future predictions regarding accident occurrence and the duration of impact. This review distinguishes itself from similar works by not only comparing models and approaches, but also by analyzing how external features, such as meteorological data, road geometric design, and land usage, affect the probability of accidents and the models' accuracy in forecasting road safety. The study explores the performance levels and limitations associated with a set of forecasting approaches, offering an analytical discussion of their differences and similarities, and potential future developments in this research space, including the use of hybrid models and reinforcement learning (RL). The results of this review indicate that DL models tend to be better suited to complex forecasting problems due to their superior ability to represent features and extract non-linear spatiotemporal correlations. This article concludes by describing various directions for further research, ranging from optimizing model architectures to integrating real-time big data into proactive prediction systems.

Keywords: traffic accident prediction; traffic data; statistical modeling; heuristic methods; reinforcement learning; machine learning; deep learning

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1. Introduction

The repercussions of road traffic accidents (RTAs) extend beyond the immediate death toll; they significantly impact economic structures and the social fabric. In addition to incurring high healthcare costs, accidents reduce productivity when individuals are killed

or incapacitated and when family members are required to leave their jobs or studies to provide care. These accidents are estimated to cost most countries up to 3% of their GDP, a figure that can climb to 5% in some developing nations, which actively hinders progress toward achieving the Sustainable Development Goals. In recognition of this serious issue, the UN General Assembly launched the "Decade of Action for Road Safety 2021–2030", with the goal of reducing road traffic deaths and injuries by 50% by 2030 [1]. According to the Global Status Report on Road Safety (GSRRS) 2023 by the World Health Organization (WHO), the leading agency for road safety within the UN, the approximate number of road traffic-related deaths reached 1.35 million people in 2021 [2]. An additional 20 to 50 million people are left with non-fatal injuries. A significant number of these casualties, over half of all road traffic deaths, are vulnerable road users, including pedestrians, cyclists, and motorcyclists. Likewise, two-thirds of all road traffic fatalities occur among individuals of working age (18–59 years), and road traffic accidents are the primary cause of death for children and young adults aged 5–29.

This global commitment shows that investing in road safety is a strategic, necessary action that positively impacts public health and the economy, making it a key part of sustainable development.

In light of increasing road safety challenges, traditional solutions like traffic signals and driving laws have proven insufficient for achieving significant and lasting improvements [3]. The fundamental flaw in these methods is their "reactive" nature; they address the consequences of accidents after they have already occurred instead of preventing them proactively.

This shortcoming has propelled research and development toward ITS, now a cornerstone of modern road safety strategies. By combining modern technologies such as data analysis and artificial intelligence (AI) algorithms such as machine learning (ML) and deep learning (DL) algorithms, these systems can predict the locations and scenarios of potential accidents [4–6]. This predictive capability goes beyond historical data, analyzing real-time and environmental factors like weather, traffic density, and driver behavior [7–9].

With this foresight, authorities can take effective, targeted preventive measures, such as adjusting traffic flow, sending instant alerts to drivers, or deploying patrols in high-risk areas. This approach effectively reduces accidents before they escalate into disasters [10]. This shift from a philosophy of "responding to accidents" to "predicting and preventing accidents" represents a crucial leap forward, which is essential for saving lives and making resource management more efficient and effective [11].

Predicting traffic accidents is no longer limited to statistical analysis, as in the past; it now involves a proactive look into the future, with the aim of saving lives before disaster strikes. This revolutionary approach relies on integrating data from multiple sources, such as driver behavior, weather conditions, traffic flow, and even vehicle data, to develop a comprehensive picture of the road environment [12,13]. Predictive models have evolved through three key stages: statistical analysis, which examines trends [14]; ML algorithms, which uncover complex patterns [15]; and, finally, DL, which handles high-dimensional data, enabling precise, real-time predictions [7]. The greatest challenge in this field is not simply developing the models but ensuring the quality and integrity of the data, which requires advanced methods to handle missing and imbalanced datasets [16,17]. This shift from a philosophy of reacting to accidents after they happen to preventing them before they occur represents a crucial leap toward safer roads for everyone.

This review builds upon the robust scientific foundations established by previous studies that addressed pivotal aspects of transport safety. The prior literature has offered in-depth analyses of the evolution of accident duration models and their associated statistical methodologies [18]. Other significant works have provided a comprehensive

understanding of injury severity by evaluating the effectiveness of ML algorithms and accurately classifying contributing factors [19]. More recent trends have explored the potential of complex DL models in predicting accidents through vehicle trajectory analysis and spatiotemporal data [20].

While acknowledging these specialized reviews, the current study offers an integrative perspective that transcends unidimensional approaches by interconnecting the trajectories of accident risk prediction, frequency, severity, and duration within a unified methodological framework. The added value of this work lies in its in-depth exploration of diverse data sources and processing mechanisms, alongside the introduction of advanced techniques previously overlooked, such as hybrid models and RL while simultaneously analyzing the rapid transition toward intelligent systems. This study distinguishes itself through an intensive focus on state-of-the-art developments between 2015 and 2025, examining 135 recent publications to identify existing research gaps. This academic maturity is further evidenced by the identification of critical "blind spots" in data analysis, reflecting a deep understanding of the gap between theoretical models and real-world operational needs. Consequently, this review provides a forward-looking roadmap, advocating for the adoption of hybrid models capable of operating effectively in dynamic environments characterized by incomplete information.

The rest of this paper is structured as follows: Section 2 describes the search methodology used in this paper. Section 3 provides background information on AI, highlighting significant concepts in ML and their distinctions. Next, Section 4 provides a discussion on data sources and data preprocessing, as well as the limitations of traditional approaches and the potential of advanced technologies in traffic accident prediction. Section 5 presents summarized ideas and considerations from an objective perspective. Finally, the conclusions of this literature review are reported in Section 6.

2. Methodology and Research Protocol

The methodology described in [21] follows a structured and rigorous approach to conducting a systematic literature review. It is primarily based on the PRISMA (Preferred Reporting Items for Systematic Review and Meta-Analysis) framework, which ensures transparency and consistency in evidence-based reporting. The process is divided into four main stages: identification, screening, eligibility, and study selection.

2.1. Search Strategy

2.1.1. Identification

In the initial phase of this study, a comprehensive literature retrieval process was conducted by searching many major electronic databases, Scopus, Web of Science, ACM, and IEEE Xplore, resulting in the successful retrieval of 330 articles. To maximize the precision and scope of the results, targeted search strings were developed using key terms such as "traffic accident", "Machine learning", "Artificial intelligence", and "data source", combined with Boolean operators (AND, OR) and wildcard symbols (*), to capture all relevant word variations. These queries were structured around three primary axes: data sources, processing techniques, and a diverse suite of data-driven computational methods. The latter encompassed statistical learning, ML, and specialized subfields including DL and RL, ensuring a robust foundation of existing research and dictionaries to guide the identification of all relevant studies (Table 1).

Table 1. Research process.

Research Data Sources	Data Processing	Research Methods
(“traffic accident” OR “traffic crash” OR “traffic collision” OR “crash risk”) AND (forec* OR detect* OR predict*) AND “data source” AND (sensor dat* OR geospatial dat* OR government dat* OR social media dat* OR open dat* OR crowdsourcing dat* OR private sector dat*)	(“data pre-process*” OR “data preprocess*”) AND (“data clean*” OR “feature extract*” OR standardi*ation OR normali*ation OR “data balanc*”) AND (“traffic accident*” OR “road accident*” OR “crash predict*”)	(“traffic accident” OR “traffic crash” OR “road traffic accident” OR “traffic collision” OR “crash risk” OR “accident duration” OR “accidents severity”) AND (detect* OR forecast* OR predict* OR estimat*) AND (algorithm* OR method* OR model OR approach OR framework OR system) AND (“statistical learning” OR “machine learning” OR “deep learning” OR “neural network”)).

2.1.2. Screening

The screening process involved two main steps to refine the initial results:

Elimination of Duplicates: There were 24 duplicated articles, but they were identified and excluded to ensure that only original records were considered for review.

Initial Filtering: The remaining 306 articles were meticulously screened based on specific inclusion and exclusion criteria (Table 2). During this step, 100 publications were excluded because they were books, book chapters, conference proceedings, or written in languages other than English. Additionally, the study’s timeframe was restricted to the last six years (2015 to 2025) to focus on current and contextually relevant literature.

Table 2. Inclusion and exclusion criteria.

ID	Inclusion Criteria
IC01	Only research that specifically used predictive modeling or artificial intelligence methods to analyze and forecast traffic accidents was included.
IC02	Peer-reviewed and indexed scientific journal papers were given priority in the review as the main sources of empirical and useful data pertaining to traffic accident prediction techniques.
IC03	In order to incorporate recent methodological advancements and developing trends in Intelligent Transportation Systems and AI-based prediction techniques, papers published between 2015 and 2025 were taken into consideration.
IC04	To ensure uniformity, accessibility, and appropriate interpretation of the reviewed literature, only English-language research were included.
ID	**Exclusion Criteria**
EC01	Duplicate records identified across multiple scientific databases were removed during the initial screening phase to avoid redundancy and duplication bias.
EC02	Studies limited to conceptual or theoretical discussions without empirical validation, experimental evaluation, or quantitative assessment were excluded.
EC03	Secondary publications, including systematic reviews, survey papers, books, book chapters, editorials, and non-peer-reviewed conference summaries, were excluded to maintain focus on primary empirical research contributions.
EC04	Studies not directly related to traffic accident prediction, predictive modeling, or Intelligent Transportation Systems were excluded, particularly those addressing unrelated analytical domains or peripheral traffic management topics without predictive objectives.
EC05	Publications that were inaccessible, incomplete, insufficiently documented, or written in languages other than English were excluded to ensure consistency, transparency, and reproducibility of the review process.
EC06	Research reporting exclusively qualitative results is excluded to maintain consistency in assessing impacts through measurable data.

2.1.3. Screening Phase

Following the screening, a comprehensive eligibility evaluation was conducted, with 206 articles identified. During this stage, the articles' titles and content were carefully reviewed to ensure their alignment with this study's research objectives. All articles provided important information. A significant number of articles, 71, were excluded at this point because they did not fit within the scope of this research or had no empirical basis, or their abstracts did not align with the aim of this study (Figure 1.)

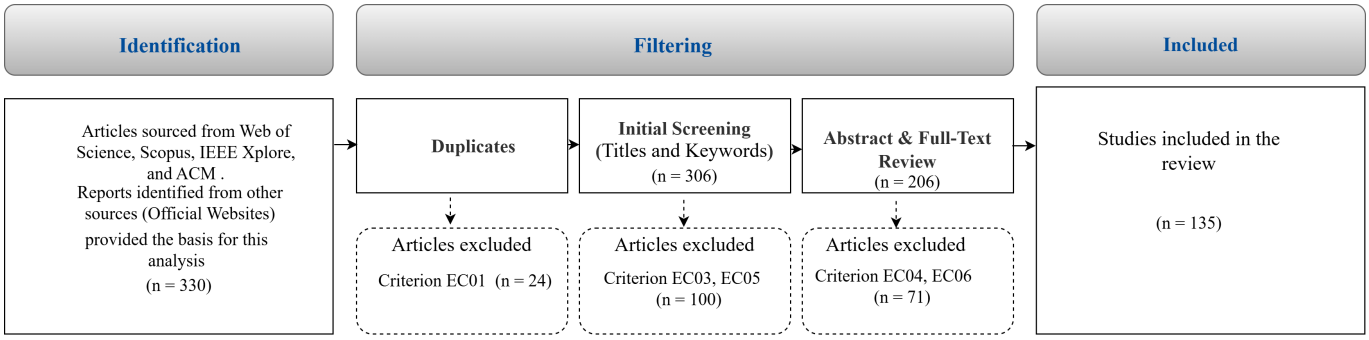


Figure 1. Identification and selection of included studies strictly adhered to established systematic review protocols.

After screening, 135 relevant articles were retained for inclusion in this systematic review.

2.2. Bibliometric Analysis

2.2.1. Keyword Co-Occurrence Network Analysis

Analysis of the keyword co-occurrence network revealed an integrated research structure that establishes ML and DL as the methodological core of contemporary road safety studies. The network highlights robust correlational links between these advanced technologies and domains such as Risk Assessment, Spatial Analysis, and the Human Factor; this reflects an academic consensus that traffic safety is a complex phenomenon influenced by both human behavior and environmental variables. Furthermore, the analysis documents a strategic integration of specific algorithms, such as random forest (RF) and support vector machines (SVMs), within applications for traffic management and intelligent transportation systems (ITSs). The convergence of these research clusters underscores that the advancement of traffic safety is fundamentally dependent on the deployment of intelligent predictive models and their capacity to process spatial and functional data with high precision (Figure 2).

This trend reflects a fundamental shift from traditional statistical modeling toward dynamic, AI-driven frameworks. The central focus on ML and DL signifies a methodological evolution aimed at capturing the non-linear complexities inherent in massive traffic datasets. These findings confirm that spatial analysis and risk assessment have become closely integrated with high-dimensional predictive algorithms to address spatiotemporal accident patterns with extreme precision. Furthermore, the prominence of these keywords suggests that intelligent transportation systems (ITSs) are evolving toward proactive management and full automation, where techniques such as GNN and LSTM provide the ability to decode human behavior and volatile traffic conditions, ensuring a real-time response that enhances overall road safety.

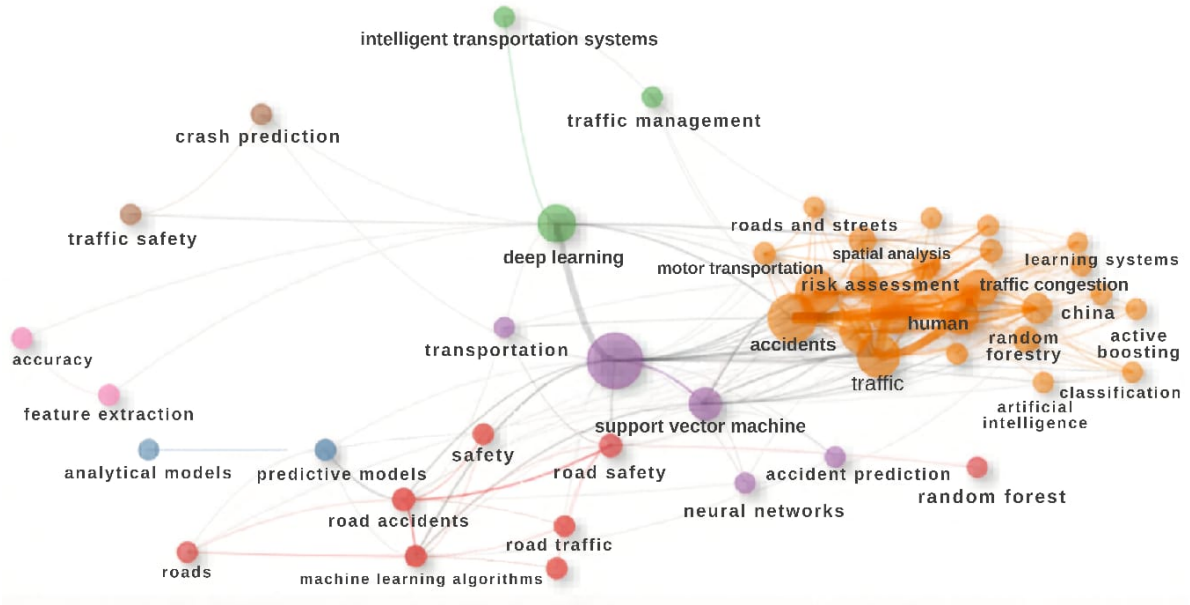


Figure 2. Keyword co-occurrence.

2.2.2. WordCloud by Title

Visual analysis of research titles within the context of intelligent transportation reflects a clear predominance of "Traffic Accidents" and "Accidents", placing transportation safety at the forefront of research priorities. The human factor emerges as a core axis with the prominence of terms such as "Human Forecasting", indicating an academic trend toward integrating human behavior into traffic prediction models. Methodologically, there is widespread reliance on AI and ML, supported by advanced algorithms such as Decision Trees and LSTM to develop robust models capable of processing complex data. Regarding environmental safety, studies focus on "Accident Prevention" and "Prevention and Control" to enhance "Car Driving" and "Road Safety". Finally, emerging and specialized research trends are driving innovation in this field, such as "Autonomous Vehicles", "Intelligent Systems", and "Object Detection", all of which accelerate the transition toward a smarter and more autonomous transportation ecosystem (Figure 3.)

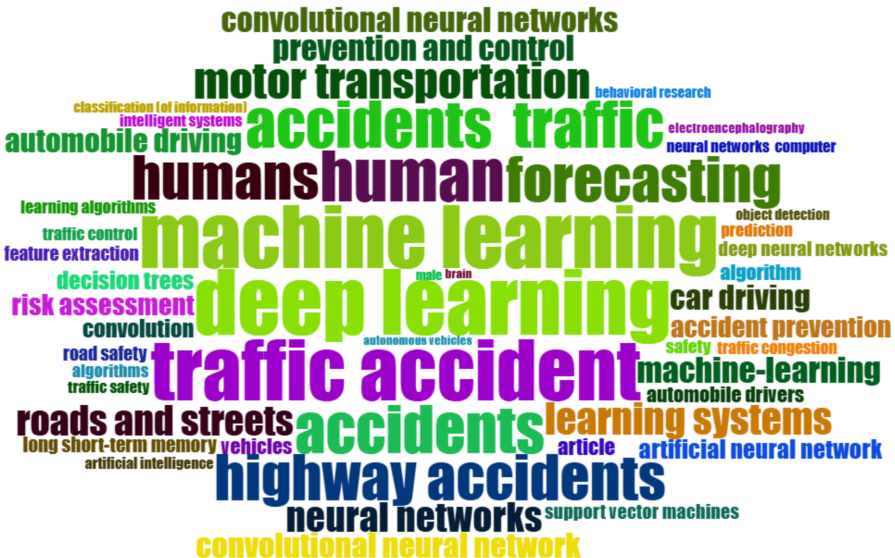


Figure 3. WorldCloud bibliometric analysis results by title.

The lexicographical dominance of DL and ML reflects a paradigm shift toward high-dimensional, non-linear predictive modeling in road safety research. The emergence of techniques such as CNN and LSTM signifies a methodological maturity focused on capturing the complex spatiotemporal dependencies inherent in massive highway accident datasets. The notable convergence between human forecasting and accident prevention underscores that future research priorities are centered on integrating behavioral dynamics and predictive analytics within ITSs. This data-driven evolution serves as the cornerstone for establishing proactive safety protocols in smart cities, where the ability to decode intricate data patterns has become a technical necessity for enhancing safety levels and traffic control in modern transportation environments.

2.2.3. Thematic Maps and Research Theme Opportunities

The thematic map shown below provides a strategic overview of the research landscape, categorizing key topics into four distinct quadrants based on their development (density) and relevance (centrality) (Figure 4).

In the upper right quadrant (motor themes) are subjects representing the most advanced and vital fields in today's landscape, characterized by their high degree of centrality and rapid technical evolution. AI, ML, DL, and big data serve as the backbone of modern research in this field. They are primarily applied in optimizing traffic flow control, enhancing road safety, and improving the accuracy of both crash prediction and feature extraction. Ultimately, this research is a fundamental driver of the ongoing development of ITSs. In the upper left quadrant (niche themes) are topics representing highly sophisticated areas of study that, while technically advanced, demonstrate limited centrality or influence within the broader general context. Often characterized by their niche specialization, these fields utilize complex models such as GRU, LSTM, and natural language processing (NLP) alongside statistical forecasting methods like negative binomial and Poisson regression. Research in this quadrant primarily explores autonomous vehicles, reward functions, and unsupervised learning. While these models are technically formidable, their impact is presently confined by narrow research silos or specialized stages of development and they have yet to achieve widespread, comprehensive application. In the bottom right quadrant (basic themes) are topics of high importance and centrality that are either still in technical development or currently serve as the fundamental pillars of the field. This domain explicitly highlights ITSs alongside traffic safety and automotive engineering. Methodologically, the focus is on data mining, feature selection, and traffic data fusion. While these subjects are essential and form the bedrock upon which current research is constructed, they require further advancement to increase their technical robustness. Strengthening these core foundations is necessary to transition them into the "motor themes" that actively drive the industry forward. Finally, the topics in the bottom left quadrant (emerging or declining themes) represent areas with a currently low degree of technical development and limited centrality within the field. This category encompasses broad domains such as general computer science and engineering, alongside more specific niches like point cloud compression. Technically, the focus of these studies remains on time series analysis and crash severity prediction. These subjects may either be nascent research trends yet to gain momentum or, conversely, general fields that are gradually being sidelined as researchers pivot toward more specialized and relevant disciplines.

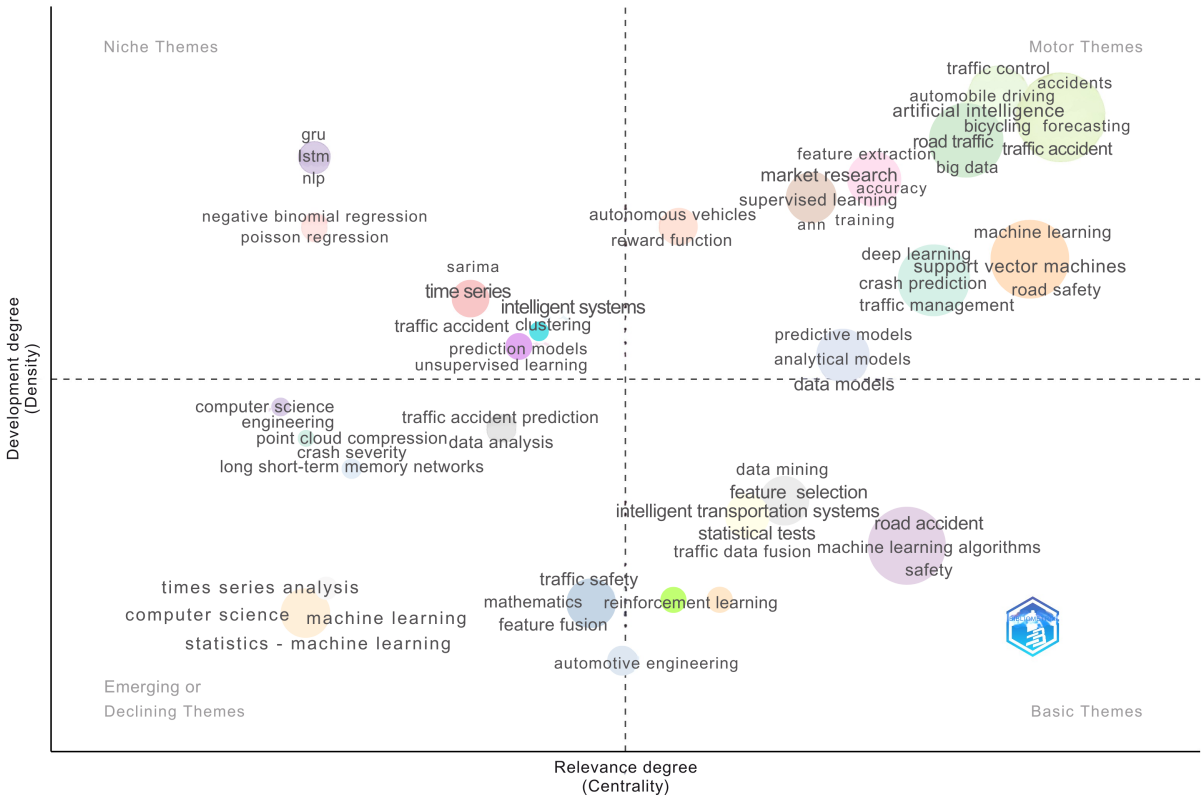


Figure 4. Results of bibliometric thematic map analysis.

On one hand, as illustrated in Figure 4, the strategic map reveals that ML, DL, and AI are the most mature and influential tools driving research in road safety and traffic management. Meanwhile, the concept of ITSs remains a foundational theme that requires further technical development to reach its full potential. Research on autonomous vehicles emerges as a technically sophisticated field; however, it continues to occupy a specialized niche rather than a central position within the broader context of general traffic and safety research. On the other hand, the central prominence of DL and AI as motor themes reflects a methodological maturity, shifting the field from traditional statistical frameworks now relegated to niche themes with limited general impact toward complex predictive modeling capable of harnessing big data complexities. The positioning of ITSs within the basic themes quadrant indicates a gap between its intrinsic importance and its technical density, necessitating future research to integrate advanced algorithms such as GNNs to address this methodological shortfall. Furthermore, emerging themes reveal a qualitative shift toward processing high-resolution sensory data, confirming that research is evolving toward proactive traffic management automation that bridges infrastructure with comprehensive intelligent analysis to ensure road safety.

3. Background: Concepts and Classifications

3.1. The General Definition of Traffic Accidents

A traffic accident may be defined as a collision event occurring on a public roadway that involves at least one moving vehicle and culminates in measurable outcomes such as property damage, personal injury, or fatality. Fundamentally, such incidents are acute disruptions to the normative operational flow of traffic, precipitated by a conflict between multiple road users or between a road user and a stationary object within the environment.

Classification of Traffic Safety Modeling Tasks

Traffic safety modeling is generally categorized into three distinct tasks based on their operational objectives, temporal focus, and data requirements.

Accident Prediction (Occurrence and Risk): This task focuses on identifying the likelihood or frequency of a crash within a specific spatial and temporal window [22]. The primary goal is prevention by proactively identifying high-risk "hotspots" or hazardous traffic states before an incident occurs. Studies often frame this as a binary classification (accident vs. non-accident) using real-time traffic flow dynamics, such as speed, volume, and occupancy, alongside weather conditions [23,24].

Crash Severity Prediction: Once an accident occurs, severity models evaluate the level of physical impact on those involved. Outcomes are typically categorized into ordinal classes: property damage only (PDO), injury, and fatal [25]. Unlike occurrence models that rely on pre-crash traffic flow, severity models prioritize human factors (driver age, sobriety), vehicle characteristics (type and speed), and infrastructural attributes (road geometry, lighting).

Accident Duration Analysis: Accident duration analysis models the time required to clear an incident and restore normal traffic flow [26]. This task is essential for incident management, helping to minimize non-recurrent congestion and secondary crashes. Most models subdivide duration into stages, such as detection, response, and clearance time. Predictive features for this task are uniquely incident-specific, including the number of blocked lanes, the involvement of large trucks, and the efficiency of emergency responding teams [24].

3.2. Common Forecasting Algorithms for Traffic Accidents

The application of computational intelligence in traffic safety has followed a distinct evolutionary path, beginning with classical ML algorithms and progressing toward more sophisticated DL architectures capable of end-to-end learning.

3.2.1. Classical ML Paradigms

ML is a fundamental branch of AI and Computer Science. It relies on the use of statistical methods to enable computers to learn, make decisions independently, and improve their performance through experience and data without the need for explicit programming [27,28]. This technology is built on the premise that computers can acquire knowledge, identify complex patterns, and derive conclusions with minimal human intervention [28]. ML can be divided into four primary branches based on the nature of the data and the training methodology:

Supervised Learning (SL)

SL relies on training models using labeled datasets [29]. This approach simulates the human learning process under a teacher's guidance, where the system learns to predict outputs based on the training examples provided to it. Supervised learning is further divided into two main categories: classification to predict specific, discrete categories (e.g., classifying emails as "spam" or "not spam") [30] and regression to predict continuous numerical values (such as market trends or weather conditions) [31].

Supervised learning algorithms like RFand SVM have been specifically compared for their effectiveness in forecasting road accidents. These models can analyze historical accident data to predict future occurrences, with RF utilizing the collective wisdom of multiple decision trees to improve accuracy and reduce overfitting in such complex datasets.

Unsupervised Learning (UL)

UL focuses on finding hidden patterns in unlabeled data, primarily through clustering [32]. Semi-supervised learning acts as a middle ground, utilizing a small set of labeled data to guide the processing of a much larger unlabeled dataset, optimizing both efficiency and accuracy. In the context of RTAP, unsupervised learning can be utilized to segment road sections based on similar safety characteristics or environmental conditions. Association rule mining can further reveal underlying correlations between different accident factors, such as weather conditions and specific road types, which may not be immediately obvious in raw data.

Reinforcement Learning (RL)

RL is a unique approach in which a software "agent" learns how to make optimal decisions by interacting with a dynamic environment. This approach is based on a trial-and-error method; the agent receives rewards or penalties based on its actions, which motivates it to refine its behavior over time to achieve a specific goal. In addition to these methods, DL has emerged as an advanced subfield [33]. It utilizes Artificial Neural Networks composed of multiple layers to model highly complex patterns in big data, representing them through various levels of abstraction [32].

For autonomous vehicles specifically, RL and DQN are vital [34], as they enable real-time decisions that prioritize passenger safety and accident avoidance. These agents progressively adapt to dynamic traffic environments, learning sophisticated behaviors to navigate complex scenarios in which traditional programmed rules might fail.

3.2.2. The Emergence of DL Architectures

DL is an advanced subfield of ML that relies primarily on Artificial Neural Networks (ANNs) designed to simulate the human brain's manner of processing complex data [35,36]. It is characterized by its ability to extract high-level patterns and abstractions from data through multiple layers of computational transformations [37–39]. Several key branches and models emerge from this field, most notably convolutional neural networks (CNNs), which have revolutionized computer vision and image recognition [40,41], and recurrent neural networks (RNNs), designed for processing sequential data such as speech and natural languages [42,43]. The latter include sophisticated architectures like LSTM and GRUs, which enhance the accuracy of processing time-dependent data [44].

For traffic accident prediction in particular, DL variants have become instrumental for enhancing safety. CNNs play a central role in autonomous driving by enabling vehicles to detect lanes and identify objects to prevent collisions in real time [45]. Additionally, LSTMs are highly effective for analyzing time-series traffic data, allowing for the forecasting of accident-prone periods based on temporal patterns in traffic flow. These models utilize their ability to capture long-term dependencies to predict future safety risks more accurately than traditional methods.

4. Analysis and Results

4.1. Road Traffic Accident Data Sources

Road traffic accident data are drawn from a complex network of sources, which collectively form an integrated information fabric for analyzing traffic phenomena. Government data and open resources represent the fundamental pillar of this system, providing researchers with documented information on accident characteristics, road infrastructure, and the surrounding environment [46,47].

In a related context, data from sensors deployed on roads, vehicles, and smart devices contribute to providing real-time and accurate readings of traffic movement [48–53].

Meanwhile, social media platforms have become a non-traditional source that provides insightful information about the number and type of accidents [54,55].

The picture is completed by geospatial data that provide an environmental context for accident analysis, recording precise details about road topography and traffic flow patterns [56,57].

Adding to this system are data provided by private sector companies, such as transportation companies and traffic control centers, as well as data provided voluntarily by the public via smart applications, which constitute supporting sources that enhance the accuracy of accident prediction [58,59].

4.1.1. Official Government and Police Databases

Most RTA research historically utilizes official accident reports compiled by police departments or Departments of Transportation (DOTs). These datasets typically contain structured information such as location, time, accident type, and severity levels [23,60]. Examples include the US-Accidents dataset, which contains over 2.25 million records [61], and the TITAN (Tennessee Integrated Traffic Analysis Network) [51].

4.1.2. Crowdsourced and Social Media Data

Real-time information is often supplemented by Volunteer Geographic Information (VGI) from mobile applications like Waze, which provides user-reported traffic alerts [51,62]. Social media platforms like Twitter and Sina Weibo serve as valuable unconventional sources, through which natural language processing (NLP) is used to extract accident and congestion information from microblogs [54,55].

These data are characterized by their high temporal accuracy and diversity of sources, making them an effective complement to traditional data [63].

Despite these advantages, there are significant challenges with the data regarding credibility, as they may contain false or repetitive information. Geographic coverage biases also appear, with data concentrating in areas of high population density [63,64]. Furthermore, there are linguistic complexities present in user-generated content.

Models based on ML and DL have proven effective in real-time accident prediction. Researchers such as Zhang, who employed DL to detect traffic accidents from social media data [62], found similar results when comparing data from police databases and loop inductors with accident reports taken from Twitter.

4.1.3. Video and Imagery Data

Advances in computer vision have made video-based data sources increasingly prominent. CCTV footage is frequently used for automated accident detection and severity classification [65]. High-resolution drone videos, such as the CQSkyEyeX dataset and the highD dataset, provide microscopic vehicle trajectory data that capture subtle kinematics and driving behaviors [66,67]. Additionally, satellite imagery and street view images are fused with traffic data to identify accident hotspots based on infrastructure and environmental context [53].

4.1.4. Sensor and Telematics Data

Infrastructure-based sensors like loop detectors provide real-time metrics on traffic speed, volume, and occupancy [68,69]. On the vehicle side, connected vehicle (CV) data [70] and digital tachographs (DTG) offer high-frequency trajectory information, enabling researchers to analyze risky maneuvers like hard braking or sudden accelerations [67,71,72].

However, the practical applicability of these models remains contingent upon data integration. Consolidating such heterogeneous sources including video feeds, textual reports, and sensor data requires a sophisticated and costly informatics infrastructure. This

necessity poses a significant barrier to implementation in resource-constrained regions, thereby limiting the global scalability of these intelligent systems [65,73].

4.2. Data Preprocessing for Traffic Accident Prediction

Data preprocessing is a critical foundation for traffic accident prediction, as it ensures data quality, consistency, and suitability for complex modeling frameworks [60,74]. The process generally encompasses data cleaning, integration, feature engineering, normalization, and handling class imbalances [22,60,75] (Figure 5).

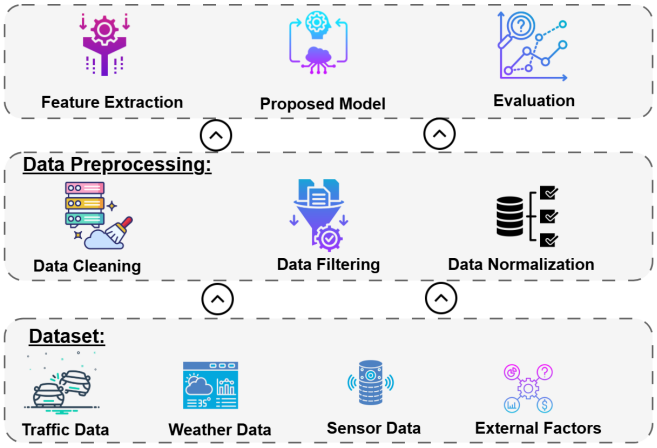


Figure 5. Block diagram.

4.2.1. Data Cleaning and Integration

The initial phase involves scrubbing raw datasets to remove incomplete, erroneous, or duplicate records [74,76]. Studies frequently filter out records with missing timestamps or locations and utilize inter-quartile range (IQR) methods to identify and eliminate outliers [60,77]. Integration is also vital, as researchers often merge heterogeneous sources, such as merging person, accident, and vehicle files using unique IDs, or fusing numerical traffic streams with unstructured text reports and weather data [75,78].

4.2.2. Feature Engineering and Selection

To enhance predictive power, raw data is transformed into sophisticated feature sets.

Text Processing: Unstructured data from accident reports are tokenized and vectorized using techniques like TF-IDF, Word2Vec, or latent Dirichlet allocation (LDA) to extract keywords and topics [78].

Spatial and Temporal Representation: The study area is often discretized into grid-based cells or "space–time cubes" to capture local dependencies [71]. Cyclic features like the "hour of the day" are often encoded using sine and cosine functions to maintain their sequential meaning [79].

Variable Screening: To reduce dimensionality and computational cost, influential variables are selected using the chi-square test, Pearson correlation analysis, or recursive feature elimination (RFE) [68].

4.2.3. Data Normalization and Encoding

Because feature dimensions vary significantly, data must be normalized to improve convergence speed. Common methods include z-score standardization, which transforms data to have a mean of zero and unit variance, and min–max normalization, which scales values to a range [66]. Categorical data is typically converted into numerical formats through one-hot encoding, label encoding, or target encoding [80].

4.2.4. Addressing Class Imbalances

A major challenge in accident prediction is that accident events are far rarer than non-accident events, leading to a biased model. To mitigate this, researchers employ resampling strategies:

SMOTE (Synthetic Minority Oversampling Technique), in which new accident cases are synthesized to balance the training set [81].

Negative sampling, in which a subset of non-accident cases are selected to match the scale of accident data [61].

Data augmentation, in which techniques like rotation, translation, and flipping are used to increase the variety of training samples for vision-based models [65].

4.2.5. Specialized Processing

For advanced DL models, preprocessing may include converting traffic features into grayscale images (FM2GI) to reveal combination relationships [74]. Other specialized tasks involve coordinate conversion (e.g., OSM to geocentric systems) and the semantic segmentation of street-view images to extract environmental features [82].

The success of accident prediction systems in smart cities depends not only on algorithmic efficiency but also on the robustness of the data infrastructure and the effectiveness of preprocessing strategies in managing the heterogeneity of multi-source data, such as connected vehicle trajectories and unstructured textual reports [67,83]. This presents a significant challenge to real-world applicability; latency in synchronizing real-time sensor streams remains the primary technical bottleneck for achieving proactive prediction at the Edge (Edge AI), which is essential for ensuring response times within windows less than 30 s [65,84]. This challenge can be attributed to the critical trade-off between increasing spatial resolution to enhance scalability and the resulting exacerbation of zero-inflation, which undermines model reliability. Consequently, the success of field deployment in ITSs is contingent upon the ability of processing frameworks to maintain a sustainable balance between data quality and instantaneous availability [79,85]. This necessitates a paradigm shift from static statistical modeling toward adaptive frameworks that address urban coverage gaps and ensure model stability through advanced resampling techniques, ultimately bolstering deployment feasibility within complex, high-dimensional traffic environments [86,87].

4.3. Traffic Accident Prediction Approaches

4.3.1. Statistical Approaches

Spatiotemporal Modeling for Accident Occurrence and Detection

High-risk location identification has seen significant advancements through the integration of geographic information systems (GIS) with advanced statistical frameworks. Studies have demonstrated that the use of the empirical Bayesian method outperforms traditional models in identifying “black spots” with high accuracy and consistency [88]. This can be attributed to the method’s ability to address the phenomenon of “regression towards the mean”, which often leads to misleading estimates in simple models. In a similar vein, research has addressed the problem of “excess zeros” and spatial heterogeneity in national road networks using spatially zero-inflated negative bounded (SZINB) models [89]. The success of this statistical approach stems from its ability to distinguish between locations that are inherently accident-free and those that have simply not experienced accidents during the observation period, providing decision makers with deeper insights compared to traditional logistic regression models [89].

Network-Wide Analysis and Frequency Forecasting

Forecasting incident frequency across regional and national networks continues to rely heavily on extensions of the Box–Jenkins method and hybrid structures. With regard to regional time series forecasting, Ihueze et al. [90] found that the ARIMAX model offers superior accuracy compared to standard ARIMA by incorporating multi-factor human and environmental explanatory variables. Similarly, in urban environments with pronounced seasonal patterns such as Belgrade, Deretić et al. [91] constructed a seasonal ARIMA (SARIMA) model to accurately capture temporal variability. Building on this strategy, Shikder et al. [92] introduced SARIMA-X and negative binomial (NB) models to analyze peak-hour intersections, concluding that, whilst NB models outperform in overall accuracy for recurrence data, multi-layer perceptron (MLP) networks are more suitable for datasets exhibiting varying event sizes.

The integration of hybrid models represents a decisive step forward in overcoming the limitations of individual frameworks, as Liu et al. [93] proposed a hybrid approach combining ARIMA and BP (backpropagation), which significantly reduced the forecast error (to 5.31%) by modeling both linear and non-linear trends simultaneously. In cases of data uncertainty or small sample sizes, Jin et al. [85] showed that Grey–Markov models outperform traditional exponential smoothing thanks to state transition probability matrices. Despite these gains, many recurrence models still suffer from unobserved heterogeneity; traditional regression models fail to address the "overdispersion" common in motorway data, a shortcoming which Abdulhafedh believes lies in a shift towards parametrized Artificial Neural Networks (ANNs) [94].

Methodological Advancements in Accident Severity and Duration

The analysis of accident severity and duration has moved from descriptive modeling to advanced structural analysis using econometric techniques. Multiple logistic regression models were used to identify the key factors influencing injury severity in urban areas, focusing on variables such as vehicle type and lighting conditions [95]. To enhance the reliability of these results, advanced statistical methods such as latent class and generalized ordered probit models with parameterized correlation structures were applied, allowing for the modeling of correlated crash outcomes and improving the robustness of accident severity estimation [96]. This ensures that the estimated coefficients are unbiased and better reflect the dynamic reality of accidents. With regard to the analysis of accident duration, risk-based accelerated failure time (AFT) models provided a detailed framework for understanding the stages of an accident (preparation, travel, evacuation) [97]. The results showed that the "gamma" distribution provides the best fit for modeling these time periods. This is primarily due to the flexibility of these models in accounting for "unobserved heterogeneity" and the influence of time-varying variables, allowing an understanding of how factors such as collision type affect each stage of the emergency response differently [97].

These studies summarized in Table 3 confirm that statistical methods remain essential for providing mathematically interpretable frameworks, something that many modern computational models lack. This interpretability stems from the reliance on well-defined probability distributions (such as the Poisson or gamma distributions). However, the major statistical gap identified is the inability of traditional models to handle complex non-linear relationships in large datasets, as well as the growing need to incorporate high-frequency, real-time traffic flow data rather than relying on aggregated annual or monthly averages [90,93,97]. The development of hybrid statistical models combining time series and spatial analysis represents the way forward for enhancing the accuracy of traffic safety forecasting.

Table 3. Taxonomy and overview of RTAP statistical approaches.

Ref	Algorithms	Prediction Task	Data Type	Key Variables	Evaluation Metrics	Strengths	Limitations
[85]	Grey GM(1,1) and Grey–Markov model	Predicting long-term trends in accidents, deaths, and property loss	National statistical yearbook data (China 2010–2017)	Historical counts of accidents, injuries, and property loss values	Absolute error and relative error (%)	Grey–Markov: Superior to basic Grey models for small/uncertain data	Standard Grey models struggle with highly volatile or rapidly fluctuating time series.
[88]	Poisson regression, NB regression, and EB method	Determining traffic accident "black spots" on highway networks	GIS-aided spatial statistical data from state roads (2005–2013)	Traffic volume (AADT), spatial coordinates, and road geometry	Accuracy, consistency, and repeatability maps	EB method: most accurate for GIS risk identification	EB outperforms NB and Poisson by providing more stable results despite data overdispersion.
[89]	SZINB and logistic regression	Simultaneous analysis of crash frequency and crash severity	Multi-level highway crash records (Thailand 2011–2017)	Segment length, lane/shoulder width, AADT, vehicle size, and drunk driving	AIC, Pseudo-R2, and Log-likelihood	Effectively addresses spatial heterogeneity and the "excess zeros" problem in crash data	Low pseudo-R2 values indicate that many influential human and environmental factors remain unobserved
[90]	ARIMA and ARIMAX (Transfer Function Model)	Forecasting monthly crash frequency	Monthly time series data (Anambra State, Nigeria 2007–2015)	Behavioral, mechanical, and environmental crash factors	Bayesian information criterion (BIC), MAPE, RMSE, and R-squared	ARIMAX outperformed by integrating complex human and environmental factors.	Monthly aggregation obscures daily patterns and limits regional generalization.
[91]	Seasonal ARIMA (SARIMA)	Forecasting future monthly traffic accident counts	Monthly time series (Belgrade, Serbia, 2016–2019)	Historical monthly accident counts and seasonal indicators	MAPE, MAE/MAD, and Theil's U1 Statistics	Highly efficient and effective at capturing seasonal variability in urban environments.	Restricted to short-term forecasting; cannot capture complex non-linear relationships.
[92]	NB, SARIMA-X, and MLP	Peak hour traffic accident forecasting at urban intersections	Hourly traffic volume and historical accident data (Regina, Canada 2015–2019)	Temporal traffic volume and intersection geometry attributes	ME, MAPE, MASE, RMSE, MAE, AIC, and BIC	NB outperformed ML and neural networks in accuracy.	Peak-hour focus causes bias, and MLPs require large diverse datasets.
[93]	Hybrid ARIMA-BP (backpropagation neural network)	Predicting all accident severities, from fatalities to property damage.	Annual historical time series (China 1990–2020)	Labor force, total population, GDP, and highway mileage	Annual average error rate, R-squared, and MSE	Hybrid ARIMA-BP: superior performance via linear and non-linear integration	Lack of real-time responsiveness and high BP tuning demands.
[94]	Poisson, NB, and ANN	Analyzing highway traffic crash frequency	5-year crash data (Interstate I-90, Minnesota 2008–2012)	Road geometry, surface condition, weather, AADT, and driver demographics	Deviance/df, Pearson Chi Square/df, and MSE	ANNs capture complex non-linear relationships much better than traditional GLMs.	NB models handle small samples poorly; Poisson models fail with overdispersed data.
[95]	MLR and ANN	Predicting the category of accident severity (Fatal/Injury vs. PDO)	Urban accident information (Rasht, Iran 2019–2020)	Accident time, geometry, weather, vehicle type, and driver age	Prediction accuracy (%), R2, and MSE	An ANN achieved 98.9% accuracy identifying high-risk vehicles and critical night hours.	Urban corridor constraint; small fatal crash data required binary categorization.
[96]	LCp-BGOP	Simultaneous prediction of injury severity for both parties involved in a crash	Empirical data of 32,308 crashes at Taipei City intersections (2018–2020)	Driver characteristics, violations, mobility mode, and crash type.	Log-likelihood, BIC, CAIC, and Likelihood Ratio Test.	Captures unobserved heterogeneity and identifies correlations between the injuries of both involved parties.	Restricted to two-party crashes and relies on initial police assessments for determining fault.
[97]	Hazard-based AFT models (Gamma, Weibull, etc.)	Analyzing factors affecting different accident duration stages (preparation, travel, clearance)	Reporting and Dispatching System (TARDS) data (Beijing 2008)	Peak hour, season, accident type, and emergency response treatment type	AIC	Allows for stage-specific variable impacts while controlling for hidden individual differences.	Lacks real-time, high-frequency traffic flow data for more granular estimation.
[98]	NBR and GPR	Modeling or predicting the number of traffic accidents in Central Java	Count data (implied by the use of Poisson and Binomial regression models for accident frequency)	Traffic accident count, adolescent involvement rate, and district/city road accident percentage.	AIC, Deviance, and Pearson Chi-Square ratios	Both models handle Poisson overdispersion, but GPR performs best with the lowest AIC (485.50).	Poisson regression yielded biased and unreliable results due to overdispersion.
[99]	Facebook Prophet, LSTM, and ARIMA	Traffic volume prediction and proactive accident prevention	IoT sensor data and road accident records (Smart city context)	DateTime, Junction ID, vehicle counts, weather, and road surface conditions	MAPE, RMSE, MAE, and Accuracy	LSTM captures temporal dependencies, while Prophet handles seasonality and missing data well.	High data sensitivity and real-time deployment complexity.

4.3.2. Heuristic Approaches

Meta-heuristic algorithms are often employed to enhance the performance of ML and DL models by optimizing their design and hyperparameters. Standard meta-heuristic techniques include Particle Swarm Optimization (PSO) [100], Genetic Algorithms (GA) [101], and Ant Colony Optimization (ACO) [102]. Such algorithms are widely employed to solve complex problems in various fields. These algorithms help navigate large search spaces to find the most efficient network configurations, which is often a challenge for standard gradient-based methods.

Spatiotemporal Modeling for Accident Occurrence and Detection

The integration of heuristic optimization within real-time surveillance frameworks has become pivotal for addressing the spatiotemporal volatility of urban traffic anomalies. Wu et al. [103] demonstrated that traditional static allocation models for incident detection often suffer from resource wastage, particularly in megacities with extensive surveillance networks. To mitigate this, they implemented a heuristic optimal scheduling approach utilizing a cyclic elimination mechanism, which reallocates computational power toward high-performing detection nodes based on historical incident density [103]. This strategy yielded a 17% to 40% improvement in detection efficiency, a success that can be attributed to the model's ability to adapt to temporal shifts in incident probability such as abnormal parking and pedestrians on highways, which are highly concentrated during specific peak windows [103]. Similarly, in the urban domain, Zhong et al. [104] addressed accident occurrence by optimizing SVM with the Sparrow Search Algorithm (SSA) to predict casualties in Wuhan. Their findings suggest that SSA-SVM effectively navigates the high-dimensional feature space of urban collisions, outperforming GA and Grey Wolf Optimizers (GWOs) by avoiding local optima during parameter selection [104]. This is mainly due to the nature-inspired foraging behavior of the SSA, which maintains a balance between global exploration and local exploitation, making it highly effective for complex urban environments where collision types (e.g., electric vehicles vs. pedestrians) vary significantly [104].

Network-Wide Analysis and Frequency Forecasting

Network screening methodologies for identifying high-risk segments have traditionally relied on data-intensive statistical models, yet recent evidence suggests that simplistic heuristic scoring can be superior in data-scarce environments. Dhakal and Al-Kaisy [105] conducted a comparative evaluation on rural highways in Oregon, contrasting a new heuristic scoring scheme against the well-established Empirical Bayes (EB) method. Their results indicate that for low-volume rural roads (AADT $\leq$ 1000), the heuristic approach demonstrates a stronger association with observed crash history, evidenced by a Spearman's Rank Correlation of 0.853 compared to 0.847 for EB [105]. This limitation in EB performance for rural domains arises from its reliance on Safety Performance Functions (SPFs) that require extensive traffic exposure data, which are often unreliable or unavailable for local agencies. Consequently, heuristic methods are most effective when used in rural network screening, as they allow for a robust ranking of potential safety improvement sites based on roadway geometry and weighted crash history without the computational overhead of complex regression [105].

Methodological Advancements in Accident Severity and Duration

The classification of accident severity has evolved through the hybridization of clustering algorithms and meta-heuristic-optimized neural networks. Alikhani et al. [106] established that the primary challenge in severity prediction is the inherent heterogeneity

of accident datasets. By employing a clustering-classification heuristic (k-means and SOM) prior to executing artificial neural networks (ANNs), they achieved higher accuracy than standalone models [106]. This enhancement can be attributed to the preprocessing stage, which partitions the data into homogeneous subgroups, thereby simplifying the non-linear decision boundaries for the classifiers.

Expanding on this optimization trend, Wang et al. [107] and Zhang et al. [108] focused on highway domains using advanced meta-heuristics to refine tree-based and neural architectures. Wang et al. [107] integrated the Crested Porcupine Optimizer (CPO) with RF to predict interstate crash severity, achieving a notable 95.2% accuracy. Their research highlights that CPO-RF is particularly effective for large-scale datasets (e.g., 110,778 records) because the CPO automates hyperparameter selection, preventing the over-fitting common in standard RF. Complementing this, Zhang et al. [108] utilized a backpropagation neural network (BPNN) optimized by Adaptive PSO for severity classification on the KABCO scale. Through Factor Analysis (FA), they identified that non-road factors, such as vehicle year and driver age, are more critical predictors than roadway geometry in highway contexts. This insight arises from the capability of the adaptive inertial weights in PSO to facilitate a faster global search, ensuring the BPNN does not become "stuck" in suboptimal weight configurations during training [108].

Collectively, the literature demonstrates that hybrid heuristic approaches—whether through prior clustering, resource scheduling, or hyperparameter optimization—provide the requisite robustness for real-world ITS applications. A prevailing pattern suggests that heuristic evaluation is optimal for data-scarce rural environments, whereas hybrid metaheuristic models are better suited for predicting incident severity in high-density urban and highway contexts. Nevertheless, a persistent limitation remains across all methodologies: the omission of real-time driver behavior data. This is a critical causal factor that historical datasets fail to capture adequately. Future research must bridge this gap by integrating driver-centric variables into these high-performance heuristic frameworks to further enhance their predictive validity.

The RTAP heuristic approaches presented in this section are listed in Table 4.

Table 4. Summary of heuristic approaches for RTAP.

Ref	Algorithms	Prediction Task	Data Type	Key Variables	Evaluation Metrics	Strengths	Limitations
[103]	Heuristic Optimal Scheduling (Cyclic Elimination)	Real-time road traffic incident detection	Surveillance camera data and incident records (Shanghai, China)	Temporal/Spatial attributes, Incident type (e.g., abnormal parking)	OETID, Load Balance Degree, Resource Utilization	17–40% improvement in detection efficiency under constrained resources	Depends heavily on prior knowledge and historical data accuracy
[104]	SVM optimized via Heuristics (GA, SSA, GWO, PSO)	Predicting traffic casualties on urban roads	Urban road traffic collision data (Wuhan, China)	Weather, Road Conditions, Collision Type, Passive/Active Hit	ACC, F1, PRE, REC, MSE, AUC	SSA-SVM effectively avoids local optima and random parameter issues	Poor performance when factors are considered in isolation
[105]	Heuristic Scoring Scheme vs. Empirical Bayes (EB)	Identifying safety improvement sites (Network Screening)	Rural highway crash history and roadway geometry (Oregon, USA)	Horizontal curve, Grade, Driveways, Speed, AADT	Spearman's Rank Correlation, RMSE, True Positive Identification	Outperforms EB for lower-volume roads; simplistic and data-efficient	EB remains superior for higher-volume road segments
[106]	ANN, ANFIS, k-means, SOM (Clustering-Classification)	Classification of road accident severity (casualties vs. damages)	Discrete traffic accident data (Iran)	Gender, road shoulder status, seat belt use, speed, road type	PCCE (Percentage of Correctly Classified Examples)	Pre-clustering significantly improves accuracy over individual models	High computational time/complexity for ensemble executions
[107]	RF optimized via Meta-heuristics (CPO, FTTA, etc.)	Motor vehicle traffic crash severity prediction	US interstate highway real-time crash data (Kaggle)	Temperature, Weather, GDP, Population Density, Time of Day	Accuracy, Precision, Recall, F1-score	CPO-RF achieved 95.2% accuracy; integrated GIS for spatial–temporal visualization	Drivers and road alignment indicators were not included
[108]	BPNN optimized by PSO with Adaptive Inertial Weights	Crash severity classification (KABCO scale)	Interstate highway crash data (Washington State, USA)	Vehicle year, Driver age, Accident type, Weather, Light	MSE, Training/Prediction Accuracy	Adaptive PSO avoids local optima; Factor Analysis provides model interpretability	High hardware requirements for BP training on big data.

4.3.3. ML Approaches

The predictive modeling of traffic accidents has fundamentally shifted from traditional econometric approaches toward ML and DL frameworks. This transition is primarily driven by the ability of non-parametric algorithms to capture complex non-linear relationships and high-dimensional interactions inherent in crash data without the rigid distributional assumptions of Poisson or negative binomial models. The recent literature emphasizes that, while global ML models provide a baseline for predictive accuracy, the integration of spatial heterogeneity, temporal dynamics, and unstructured data is essential for robust safety management across different tasks and domains.

Accident Occurrence and Real-Time Risk Identification

The task of predicting accident occurrence has evolved from static hotspot analysis to high-resolution, real-time risk identification in both urban and highway domains. Several studies [79,109] have demonstrated that ensemble learners, such as RF and XGBoost, are particularly effective in dense urban environments when coupled with space–time cubes to reduce false discovery rates. This success can be attributed to the models' robustness against noisy datasets and their capability to process heterogeneous data sources, including road geometry, weather, and historical counts [79,109].

In highway scenarios, effectiveness often relies on the fusion of macro-level traffic flow and micro-level kinematics. Huang et al. [67] showed that joint approaches leveraging vehicle trajectories from drones significantly outperform single-source models, with XGBoost achieving an AUC of 0.976. For time-critical events like hurricane evacuations, Syed and Hasan [84] reported that Gaussian process boosting (GPBoost) achieved high recall rates (0.91) by leveraging connected vehicle (CV) data. This effectiveness arises from the capability of boosting architectures to handle the extreme class imbalance (rare events) typically found in real-time safety data [67,84]. However, a notable limitation arises from the heavy dependency on high-frequency data streams, which may not be universally available in all roadway segments.

Accident Frequency and Spatial Heterogeneity

Accident frequency estimation increasingly utilizes spatial ML to address the limitations of traditional "global" models that ignore localized variations in risk factors. Researchers [110,111] investigated geographically weighted random forest (GW-RF) and geographic support vector regression (GSVR) to model zonal-level crash counts. Wang et al. [110] demonstrate that GW-RF significantly outperforms global RF and ordinary least squares (OLS) models by identifying how specific variables, such as intersection density or land use, vary in influence across different geographic regions.

In urban domains, the inclusion of micro-level driving behaviors such as hard braking and rapid turning extracted from CV trajectories has been shown to improve intersection frequency prediction. Han and Abdel-Aty [112] developed a spatial ML framework using LightGBM, noting that while downtown crashes are primarily driven by hard braking (rear-end risks), suburban intersections are more sensitive to lateral turning maneuvers. This limitation in global models arises from the assumption of spatial stationarity, which fails to account for the diverse sociodemographic and environmental contexts of a city [110,112].

Accident Severity and Model Interpretability

Injury severity prediction remains a critical task, often hampered by severe class imbalance where fatal accidents represent a minute fraction of the data. To mitigate this, studies [113,114] proposed multi-layer stacking frameworks and ensemble structures.

Tang et al. [113] demonstrated that a stacking framework integrating RF, AdaBoost, and GBDT achieves superior accuracy and recall in predicting injury levels at freeway diverge areas compared to standalone SVMs. Similarly, Krishna et al. [115] and Katoch et al. [116] report that tree-based models and neural networks consistently outperform logistic regression in multi-class injury classification.

A recurring pattern across severity studies is the shift from "black box" prediction to interpretable AI. Tang and Wang [117] developed an improved LightGBM method utilizing CU_SMOTE and SHapley Additive exPlanations (SHAP) to identify critical factors like safety device usage and road type. Islam et al. [118] further highlight that decision trees (CART) provide more convenient and accurate interrelation insights for specific demographics (e.g., productive age groups) than logistic regression. This trend is mainly due to the need for traffic agencies to derive actionable safety policies such as targeted enforcement for driver distraction from model results [60,118].

Accident Duration and Unstructured Data Integration

Predicting accident duration is essential for highway incident management and congestion mitigation. The recent literature [60,83] has investigated the integration of unstructured data, such as text mining from incident reports, with traditional structured data. Chen et al. [83] demonstrated that a hybrid TF-IDF and RF approach significantly improves duration prediction on expressways by capturing textual descriptors of the scene. This is mainly due to the fact that text data often contains nuanced situational information (e.g., vehicle fire or hazardous materials) not captured in binary traffic logs [83].

In specialized scenarios, such as geohazard-induced accidents on mountain roads, Bai et al. [119] show that random survival forest (RSF) models outperform traditional Cox proportional hazards regression by better capturing non-linear relationships between geomorphology and clearance time. For rural interstate clearance, Khan et al. [60] report that XGBoost is superior for intermediate and long durations. However, a persistent limitation arises from the lack of standardized reporting in text data, which complicates the feature engineering process for regional-scale applications [83].

In summary, ML performs best in classification tasks (severity) and frequency estimation where static roadway, environmental, and behavioral factors interact in complex, non-linear ways. It is particularly robust in urban environments when augmented with spatial weighting (GW-RF, GSVR) or stacking techniques. While Hybrid methods are bridging the gap toward higher accuracy, ML models remain the benchmark for accident duration and severity due to their lower computational cost and superior interpretability through XAI tools. The shift toward incorporating unstructured text data and spatial non-stationarity signifies a move toward models that are not only more accurate but also more transparent for safety interventions in both urban and highway domains. Refer to Table 5 for a summary of the RTAP ML strategies discussed in this section.

Table 5. Overview of RTAP ML models.

Ref	Algorithms	Prediction Task	Data Type	Key Variables	Evaluation Metrics	Strengths	Limitations
[25]	MNL, NNC, SVM, RF	Crash severity prediction (4 levels)	Structured crash records	Driver age, road type, geometry, lighting	Correct prediction rate, OPE, SPE	NNC with K-means effectively predicts severe crashes.	MNL performs poorly with imbalanced datasets
[60]	XGBoost, CatBoost, LightGBM	Incident clearance time (ICT)	Rural interstate crash database	Crash severity, heavy trucks, driver age, weather	Accuracy, precision, recall, F1	XGBoost excels at nonlinear duration prediction.	Often treated as "black boxes"; SHAP improves interpretability.
[67]	LR, kNN, XGBoost, RF, MLP	Highway risk event identification	Vehicle trajectory (HighD) and flow	Minimum distance (Min_D), acceleration	F1-score, AUC	Synergistic use of flow and inter-vehicle kinematics surpasses single features	SMOTE and RENN show metric-dependent performance variability.

Table 5. *Cont.*

Ref	Algorithms	Prediction Task	Data Type	Key Variables	Evaluation Metrics	Strengths	Limitations
[75]	ANN-MLP, SVC, NBC, DT, CBR	Injury severity (fatal vs. non-fatal)	Structured accident records (NHTSA)	Belt usage, alcohol consumption, speed violation	Accuracy, precision, recall, F1-score	MLP achieved highest accuracy; SHAP used to explain black box behavior	Complexity obscures direct causal interpretation without auxiliary tools.
[76]	SVM, GPR, ANN, DT, Ensemble Trees	Freeway incident duration (continuous)	Large-scale incident dataset	Weather, vehicle type, intensity, month	MAE, RMSE	SVM and GPR achieved the best MAE (14.34 min)	ANN and SVM require significantly longer training times
[77]	SVM, kNN, RF, MLP, GPC, DT, LR, NB, DA, SGD, BE	Freeway incident duration classification	Houston TranStar records	Severity, number of vehicles, lane blockage	Accuracy, Precision, Recall, F1, AUC, Time	SVM and kNN achieved 97% accuracy; SVM is memory efficient	DT recorded lowest performance (82%); GPC is computationally slow
[79]	XGBoost	Crash risk hotspot identification	GIS space–time cubes	POI density, land use, parking meter density	Accuracy, precision, recall, F1, AUC	Effectively captures spatiotemporal correlations and neighboring features	High computational demand for large-scale urban spatial matrices
[83]	TF-IDF-RF, GBDT, XGBoost	Accident duration prediction	Unstructured text + structured data	Report text keywords (fire, rollover), hour	RMSE, MAPE, R-squared	Text mining identifies critical qualitative factors missed by structured data	Sparsity of text data and high dimensionality of keywords
[84]	XGBoost, GPBoost, LR, SVM, RF	Real-time crash risk (hurricane evacuation)	Connected vehicle (CV) micro-data	Upstream vehicle count, downstream mean speed	Recall (sensitivity), F1, accuracy	GPBoost accounts for spatial correlations; high recall (0.91)	Small dataset size (3 days of evacuation) limits generalization
[109]	BRF	High-resolution collision prediction	City-wide open data (Montreal)	Visibility, count of past accidents, temperature	Recall, false positive rate (FPR)	Spark-based BRF handles severe class imbalance efficiently	High resolution leads to many "false positives" that are actually high-risk scenarios
[110]	GW-RF	Zonal-level crash frequency	Multi-source spatial data (US)	Motorway density, intersection density, poverty	R-squared, MSE, AIC	Addresses spatial non-stationarity better than global RF or GWR	Model performance varies significantly by region/zone type
[111]	GSVR	Zonal-level crash frequency	Travel demand (MABM) and GIS data	Traffic volume, VKT, land use entropy	MAE	Incorporates generated distance matrices to capture unique spatial dynamics	Relies on simulated data from MABM which may introduce noise
[112]	Spatial RF, XGBoost, LightGBM	Intersection crash frequency	CV and GSV (Google Street View) images	Hard braking, radial acceleration, GSV pixels	RMSE, MAE, R-squared	Incorporates visual environment and risky driving behavior (longitudinal/lateral)	Requires high-resolution CV data and complex pixel segmentation
[113]	Stacking (RF, AdaBoost, GBDT) with LR	Crash injury severity classification	Freeway segment data	Deceleration lane length, mainline ADT, ramp ADT	Accuracy, recall	Two-layer stacking framework outperforms individual base classifiers	High number of hyperparameters requires complex calibration
[114]	RF, DT, SVM, kNN, ANN, NB, LR, GB	Accident severity classification	Large-scale road accident data	Road type (highway), speed, daylight, gender	AUC, accuracy, precision, recall	RF demonstrated superior predictive performance for fatal vs. non-fatal crashes	Recall for fatal incidents remains a challenge due to rarity
[115]	DT, RF, logistic regression	Road accident prediction and analysis	Historical accident dataset	Speed limit, age, weather, vehicle type	Accuracy	RF achieved the highest accuracy (88.89%)	Accuracy is highly dependent on proper sample selection/cleaning
[116]	NN, RF, DT, AdaBoost, SVM, NB, kNN, SGD, LR	Road accident severity classification	Aggregated Indian traffic data	Weather, road infrastructure, demographics, location	Accuracy, precision, recall, F1-score	Neural network outperformed all others with 90% accuracy	Limited by single-area data and potential underreporting of minor incidents
[117]	Improved LightGBM	Accident severity prediction	Imbalanced traffic accident data	Safety equipment (P-SAFE), collision type, road type	Accuracy, recall, F1-score	CU-SMOTE and focal loss effectively address class imbalance and severe accidents	Requires visual SHAP analysis to make findings actionable for safety policy
[118]	Binary LR, CART	Young driver crash involvement (15–44 yrs)	Regional crash data (Saudi Arabia)	Speeding, distraction, improper turning, gender	Accuracy, precision, recall, F1	CART provides better interpretability for inter-variable relationships	Logistic regression assumes linearity that may not hold in complex behavior
[119]	Random survival forest (RSF), CPH, survival SVM	Geohazard accident duration	Text data and survival (censored) data	Disaster type, secondary accidents, morning peak	C-index, AUC, MAE, MSE	RSF handles censored data and non-linear geohazard impacts superiorly	Traditional models (CPH) rely on flawed proportional hazard assumptions

Table 5. *Cont.*

Ref	Algorithms	Prediction Task	Data Type	Key Variables	Evaluation Metrics	Strengths	Limitations
[120]	DT, LightGBM, XGBoost	Accident severity (fatal, serious, slight)	UK 2020 road accident data	Vehicle type, maneuver, number of vehicles, age	Accuracy, precision, recall, F1, specificity	DT with default parameters provided reliable baseline results	Performance impacted by limited travel under COVID-19 restrictions
[121]	RF	Road accident prediction and safe routing	GIS and historical accident data	Road features, traffic conditions, time, location	R-squared, RMSE	Integrated spatial network analysis identifies safest paths based on risk	Relies on historical data which may not reflect real-time variability
[122]	RF and dynamic Bayesian network	Highway traffic crash risk prediction	Temporal traffic flow data	Mean speed, upstream occupancy, speed difference	Accuracy	DBN captures dynamic time-varying characteristics better than static models	Accuracy varies significantly across different time segments

4.3.4. Reinforcement Learning Approaches

The Shift from Static Predictive to Dynamic Proactive Frameworks

The landscape of traffic accident research has undergone a significant paradigmatic shift, moving from traditional ML architectures toward goal-oriented RL frameworks. This transition is mainly due to the inherent inability of static ML models, such as RF and standard DL, to adapt to the stochastic feedback loops of real-time traffic [26]. For instance, while classical ML remains effective for initial risk quantification and establishes high accuracy in predicting vehicle numbers in rural crashes, its performance often plateaus in dynamic environments. This can be attributed to the lack of an online learning mechanism that allows the system to adjust its policy based on the immediate outcome of an intervention [123]. Consequently, deep reinforcement learning (DRL) has emerged as a superior method for integrating accident occurrence and frequency prediction directly into active management strategies, such as adaptive traffic signal control (ATSC) and variable speed limits (VSL) [26,123].

Proactive Safety Management in Highway and Freeway Domains

Within the highway domain, the effectiveness of RL is most pronounced in mitigating risks associated with traffic oscillations and tunnel environments. Research by Jin et al. [124] and Li et al. [125] demonstrates that DRL agents, specifically those utilizing Deep Q-Networks (DQN), perform exceptionally well when tasked with real-time safety perception and VSL optimization. These models are effective when traffic flow exhibits stop-and-go behavior, where they can proactively smooth density waves before a crash occurs [125]. This superior performance arises from the integration of macroscopic models, like the cell transmission model (CTM), with dynamic Bayesian networks (DBN) to capture hazard patterns that traditional regression models fail to identify [123]. However, a critical limitation in this domain is the heavy reliance on high driver compliance assumptions; if drivers ignore speed limit changes, the safety benefits are markedly reduced. Furthermore, while RDCN-based perception modules capture unobserved heterogeneity in tunnels [124], they remain sensitive to the accuracy of the underlying traffic flow simulations used during the training phase [124].

Urban Resiliency and Intersection Safety Optimization

In complex urban domains, RL frameworks focus on enhancing accident severity management and infrastructure resiliency through signalized intersection control. Zeinaly et al. [126] and Chan et al. [26] highlight that DRL agents, such as those employing DQN and experience replay, are highly effective during disruptive accident scenarios. These systems are designed to be resilient, meaning that they can manage traffic flow and reduce queue lengths even when the location or time of an accident changes, without requiring extensive retraining [126]. This can be attributed to the agents' ability to generalize

traffic states across discretized lane cells rather than relying on fixed detector points. A significant advancement in this area is the RECOSAM model, which quantifies safety as a monetary cost, allowing for a direct, justifiable trade-off between traffic delay and accident severity [26]. Despite these strengths, a primary limitation arises from the computational overhead of integrating high-fidelity safety evaluation tools (like SSAM) into real-time DRL loops, which often necessitates the use of simpler surrogate models for training [26].

Perception Robustness and Explainability in Autonomous Systems

Recent studies have extended DRL to the domain of autonomous vehicles (AVs) and collaborative network environments to address accident avoidance and detection. Muzahid et al. [127] and Han et al. [128] demonstrate that actor-critic methodologies (e.g., PPO, SAC, DDPG) excel in managing multi-agent tasks such as chain collision avoidance and urban lane-changing. These models are effective where high-dimensional sensor data (LiDAR, Video, Radar) must be fused to ensure safe maneuvering [127,129]. To solve the "black box" logic inherent in deep architectures, the integration of fuzzy logic with RL has been shown to improve the interpretability and anthropomorphism of driving decisions, making them more acceptable to human drivers [128]. Furthermore, the SMART mechanism enhances detection robustness in noise-heavy urban networks by utilizing self-supervised learning to prioritize core safety semantics [129]. This robustness is mainly due to the use of Transformer-GNN fusion, which preserves critical traffic data even during high packet loss. Nevertheless, the limitation of transferability persists, as these algorithms are largely validated in 3D simulation environments like CARLA or Unity3D, leaving a "reality gap" for large-scale real-world deployment.

The synthesis of the current literature reveals a clear trajectory toward multi-objective hybrid RL frameworks that unify perception, prediction, and intervention. While ML remains the standard for offline risk assessment and cost quantification, DRL is essential for active, real-time safety interventions in both urban and highway settings. The pattern across these studies suggests that the most successful models are those that integrate multimodal sensor fusion with explainable logic (like fuzzy logic) or economic safety metrics (monetary costs). A recurring critical gap is the lack of cross-validation in geographically dispersed real-world networks and the exclusion of human factors like low driver compliance and weather-induced uncertainty in training scenarios. Ultimately, bridging the gap between high-fidelity simulation and real-world heterogeneity remains the primary challenge for the next generation of safety-critical intelligent transportation systems.

The studies noted in Table 6 demonstrate a significant shift toward integrating DRL with real-time predictive modeling to enhance urban and rural transportation safety [123,124]. A primary trend across this literature is the transition from single-objective optimization (e.g., merely reducing travel time) to multi-objective frameworks that balance traffic efficiency with quantifiable safety metrics, such as crash probability or monetary safety costs [26,124]. Furthermore, the adoption of multimodal data fusion (Radar, LiDAR, Video) and self-supervised learning highlights a growing emphasis on system robustness and communication efficiency in complex network environments [129]. While these studies report substantial improvements in accident resilience and collision avoidance, a recurring limitation is the reliance on high-fidelity simulation environments (e.g., SUMO, CARLA, Unity3D), suggesting a critical need for future research to validate these anthropomorphic and adaptive algorithms in diverse, large-scale real-world deployments [124,127–129].

Table 6. Summary of RL approach for RTA forecasting.

Ref	Algorithms	Prediction Task	Data Type	Key Variables	Evaluation Metrics	Strengths	Limitations
[26]	RECOSAM (XGBoost) + Deep RL (DNN)	Preemptive crash risk reduction in ATSC	Simulated intersection traffic (Vissim)	Traffic volume, waiting time, and turning proportions	Safety cost (monetary), delay, and queue length	Quantifies safety as a monetary cost, enabling direct trade-offs with efficiency	The full predictive capabilities of the safety model were not fully realized
[123]	DQN and dynamic Bayesian network (DBN)	Proactive road safety using variable speed limits (VSLs).	Expressway traffic data (Tokyo Metropolitan Expressway)	Traffic flow, average speed, and occupancy	Crash likelihood, overall accuracy, and number of crash-prone events	Successfully combines risk prediction and intervention via a Cell Transmission Model (CTM).	Assumes 100% driver compliance with speed limits
[124]	MDRL + RDCN	VSL control for freeway tunnels with safety perception	Simulated tunnel traffic (SUMO) and historical crash data	Traffic flow, occupancy, speed, and weather conditions	Traffic conflicts, time-loss, and crash probability	Balances safety and efficiency while reducing frequent speed limit changes via a safety trigger	Performance relies on the accuracy of simulation-based training
[125]	Q-Learning	VSL control to reduce crash risks near traffic oscillations	Aggregated traffic variables in CTM simulation	Traffic density and speed limit values	Total crash risk (TCR) and total travel time	Effective continuous online learning to handle real-world uncertainties	RL agent used limited and discontinued states for traffic density
[126]	DQN with experience replay	Adaptive traffic signal control (ATSC) during accidents	Microscopic traffic simulation (SUMO)	Vehicle presence, queue length, and signal phase	Queue length, waiting time, and cumulative reward	Resilient to disruptions; does not require retraining if accident locations change	Limited to a single four-way intersection scenario
[127]	PPO, SAC, and DDPG (actor–critic)	Chain collision avoidance for autonomous vehicles (AVs)	Simulated AV traffic in 3D environments (Unity3D)	Vehicle velocity, relative distance, and position	Success rate, collision frequency, and computational cost	Comprehensive comparison of multi-agent and single-agent efficiencies	Did not fully incorporate weather or road profile uncertainties
[128]	Fuzzy logic + deep deterministic policy gradient (FL-DDPG)	Urban lane-changing decision-making	Simulated urban traffic (CARLA)	Ego-vehicle speed, relative speed, and safety distance	Average reward, collision rate, and travel efficiency	Anthropomorphic decision-making and improved explainability through fuzzy logic	Neglects the braking effect in the current decision model.
[129]	Self-supervised multimodal and RL (SMART)	Multimodal traffic data transmission and semantic processing	Multimodal (Video, Radar, LiDAR)	Bandwidth, latency, packet loss, and semantic similarity	Semantic success rate (SSR), latency, and F1-score	Robustness to noise and high transmission efficiency under bandwidth constraints	Not yet validated on large-scale geographically dispersed networks

4.3.5. DL Approaches

The evolution of traffic safety research has shifted from traditional statistical paradigms toward complex DL architectures, a transition necessitated by the highly non-linear and spatiotemporal nature of transportation data. Recent scholarship indicates that, while classical ML models like RF and SVM provide strong baselines, they often fail to capture the intricate dependencies inherent in modern ITSs [22,23,87].

Spatiotemporal Modeling for Accident Occurrence and Detection

The superior performance of architectures such as LSTM and gated recurrent units (GRUs) in accident occurrence and real-time detection tasks is mainly due to their ability to model long-range temporal dependencies [22,69,130]. This is particularly evident in highway domains, where models must process high-frequency loop detector data to identify the rapid transitions in traffic flow that precede a crash [22,69]. Research by Jiang et al. and Parsa et al. [22,130] highlights that LSTMs excel in this domain because they mitigate the vanishing gradient problem, allowing the model to retain crucial pre-crash traffic signatures over multiple time steps. Furthermore, the effectiveness of these models is enhanced when utilizing multiple temporal resolutions (e.g., 5-min and 30-min intervals), which allows the system to capture both immediate volatility and broader traffic trends [22].

In urban domains, the challenge shifts toward capturing simultaneous spatial and temporal correlations across complex road networks. The adoption of convolutional long short-term memory (ConvLSTM) and hybrid CNN-LSTM architectures (such as STCL-Net) has proven highly effective for citywide short-term crash risk prediction [23,82]. This can be attributed to the CNN component's ability to extract topological features from grid-

based urban representations, while the LSTM layers process the temporal evolution of risk. These models perform best in high-density urban environments where accident clusters are influenced by localized factors like land use and secondary incidents [23]. However, a persistent limitation arises from the "zero-inflated" problem in high-resolution urban grids, where the rarity of accidents compared to non-accident states can lead to model bias if not addressed through sampling techniques like SMOTE [23,69,74].

Network-Wide Analysis and Frequency Forecasting

For tasks involving accident frequency and large-scale incident analysis, there is an emergent trend toward graph neural networks (GNNs) and dynamic state-space models [87,131]. Recent findings demonstrate that GNNs outperform Euclidean-based models by treating the road network as a non-Euclidean graph, which more accurately represents real-world infrastructure [131]. The success of GNNs in this context is mainly due to the integration of "concurrency priors", which acknowledge that an incident at one node significantly increases the probability of incidents at neighboring nodes. Similarly, the use of denoising autoencoders (DAEs) for unsupervised feature learning, combined with dynamic state-space models, has shown superior performance in estimating crash likelihood [87]. This approach is particularly effective for accommodating unobserved heterogeneity in correlated crash data, which traditional regression models often overlook.

Methodological Advancements in Accident Severity and Duration

Research focusing on accident severity classification has moved toward multi-task learning and vision-based architectures to overcome the limitations of single-task discrete outcome models [69,74,132]. Multi-task deep neural networks (DNNs) are reported to be superior for severity prediction because they leverage shared structural information between related outcomes (e.g., injury, death, and property loss) [132]. In the domain of vision-based surveillance, 19-layer CNNs have been deployed to classify severity directly from CCTV footage, achieving high accuracy by learning hierarchical features from raw pixels [69]. Furthermore, innovative approaches like TASP-CNN transform tabular accident data into gray images, enabling the CNN to "see" combination relationships between features that are often invisible to traditional ML models [74].

Regarding accident duration, social media data has emerged as a vital secondary source for detection and impact assessment [62]. Deep belief networks (DBNs) used for processing accident-related tweets have shown that nearly 66% of such reports on freeways can be tied to official accident logs. The effectiveness of social media in this task is attributed to its ability to capture "unrecorded" minor incidents that may not trigger infrastructure sensors but still impact traffic flow. However, the limitation arises from the noisy and unstructured nature of such data, requiring intensive natural language processing and feature selection to maintain precision [62].

The synthesis of these studies reveals that DL architectures, particularly spatiotemporal models (ConvLSTM) and graph-based networks (GNNs), represent the current gold standard for traffic accident prediction. Their success is contingent upon the availability of high-resolution temporal data and the integration of multi-source data (e.g., POI, weather, and social media). While urban domains benefit most from grid-based or graph-based spatiotemporal modeling, highway domains require high-frequency temporal architectures focused on flow transitions. Future research must bridge the gap between predictive accuracy and model interpretability to ensure that these systems can be effectively utilized by transportation authorities for proactive safety management. A synthesis of the RTAP DL strategies analyzed in this section is displayed in Table 7.

Table 7. Summary of the discussed RTAP DL strategies.

Ref.	Algorithms	Prediction Task	Data Type	Key Variables	Evaluation Metrics	Strengths	Limitations
[22]	LSTMDTR (LSTM with different temporal resolutions)	Crash detection (binary classification)	Freeway traffic sensor data (California)	Speed, flow, occupancy	Overall/crash accuracy, false alarm rate (FAR)	Captures both long-term and dynamic short-term pre-crash conditions	Increased neurons/resolutions lead to higher computation time
[23]	STCL-Net (CNN, LSTM, ConvLSTM hybrid)	Citywide short-term crash risk	Multi-source data (GPS, road, land use, weather)	Taxi GPS data, road network, population, land use, weather	MSE, MAE, $MAPE^H$	Superior at capturing spatiotemporal characteristics from multi-source data	Model performance decreases as spatiotemporal resolution increases
[61]	DAP (deep accident prediction) with embedding	Real-time accident risk prediction (15-min intervals)	Large-scale heterogeneous sparse data (US-Accidents)	Traffic events, weather, points-of-interest (POIs), time	F1-score	Uses easy-to-obtain sparse data; scalable for citywide prediction	Missing records in streaming data can impact performance
[62]	DBN, LSTM	Binary classification detecting traffic accidents.	Unstructured tweets validated using official accident logs and 15,000 loop detectors.	44 unigrams and 17 bigrams as binary accident indicators.	Overall accuracy and precision	Achieves 85% accuracy, detects unrecorded minor accidents, and outperforms traditional sensors in reporting speed.	Noisy unstructured data, sparse geo-tags, map-matching errors, and false-alarm risk.
[65]	19-layer CNN	Accident detection and severity (4 levels)	CCTV footage (video/images)	Visual pixel data from surveillance; contextual info (location, date, time)	Training/validation accuracy (98.17–99.04%), AUC (0.95)	High accuracy in real-time detection; reduces human error/subjectivity	Dependency on CCTV infrastructure; performance varies with lighting/weather
[68]	CNN with chi-square test for variable selection	Accident occurrence likelihood (30 min ahead)	Spatiotemporal expressway data (Japan)	Speed, traffic volume, occupancy, weather, day of week, time of day, accident interval	F1-score (improved by 19.1%)	Adds explanatory variables (weather/time) to basic traffic data; short prediction window	Specific to expressways; requires detailed sensor data (detectors)
[69]	LSTM, GRU with SMOTE for balancing	Real-time accident detection	Spatiotemporal sequential data (Chicago)	Speed, volume, occupancy (upstream/downstream), weather, peak hours	Accuracy, detection rate, false alarm rate, AUC (0.85)	Effectively handles imbalanced data via SMOTE; good real-time performance	Computationally more expensive than traditional models; requires loop detector data
[74]	TASP-CNN; FM2GI algorithm	Severity prediction (slight, serious, fatal)	8-year historical tabular data converted to red images	Crash data maps location, road class, time, vehicle count, surface, weather, casualties, and vehicle types.	MicroF1, precision, recall	Successfully excavates latent feature representations and feature correlations	Image conversion process (FM2GI) adds complexity; limited to historical records
[82]	ConvLSTM (Convolutional LSTM)	Crash prediction (spatial and temporal)	Grid-based historical crash data	Historical crash patterns mapped to redscale images	Cross entropy, MSE, RMSE, CSI, FAR, POD	Successfully captures spatial characteristics within road networks	Difficulty in collecting large-scale, high-quality data in developing countries
[87]	DAE (denoising autoencoder) + multivariate tobit	Crash estimation and occurrence prediction	Intersection-level multivariate data	Traffic (AADT), geometry, pavement condition, environment	RMSD, MAE	Accounts for unobserved heterogeneity; improves estimation accuracy	Requires detailed roadway asset management data
[130]	RNN, LSTM, CNN+LSTM, GRU, Transformer, MLP	Accident count forecasting (yearly/monthly)	10-year time-series database (India)	Historical accident counts, weather, road infrastructure, traffic volume	RMSE, MAE, MAPE	Comprehensive comparison of DL architectures; LSTM/GRU showed superior performance	Lacks spatial detail; focuses on counts rather than individual accident characteristics
[131]	GNNs (GCN, GraphSAGE, etc.) with concurrency prior (CP)	Semi-supervised traffic incident prediction	Large-scale nationwide graph-based road data	Road type, max speed, lanes, average neighbor crash density (ANCD)	F1-score, AUC	Statistically validates concurrency hypothesis (neighboring incidents); scalable	Complex graph construction; sensitive to labeling sparsity
[132]	Multi-task DNN with layer-wise relevance explanation (LRP).	Multiple severities (injury, death, property loss)	Large-scale tabular data (China)	Driver demographics, vehicle characteristics, road geometry, and environmental conditions.	Accuracy, AUC	First multi-task model for comprehensive severity; high explainability via LRP	Model complexity increases with the number of tasks; black box nature of DNNs without LRP
[133]	Deep LSTM networks; empirical CDF	Predictive pedestrian collision risk prevention.	CCTV video footage (unsignalized crosswalks)	Vehicle/pedestrian trajectories (position, speed, angle)	MSE: trajectory; danger, warning, caution levels	Proactive (not post facto); addresses vulnerable road user (VRU) safety	Limited to "interactive scenes"; trajectories can be "zigzag" due to camera noise

4.3.6. Hybrid Approaches

The transition from traditional statistical methods to complex hybrid DL architectures marks a significant evolution in traffic safety analytics, driven by the need to capture high-dimensional spatiotemporal dependencies and heterogeneous data sources. This synthesis evaluates contemporary research across varying tasks and domains, emphasizing the synergistic benefits of integrated modeling approaches.

Spatiotemporal Modeling for Accident Occurrence and Detection

The superior performance of models such as the deep spatiotemporal hybrid network (DSHN) and CNN-DNN in predicting real-time accident occurrence can be attributed to their capacity to extract localized spatial features while maintaining temporal continuity. Expanding on this hybrid strategy, Kashifi et al. [73] presented a hybrid architecture that integrates convolutional layers to capture the spatial heterogeneity of urban road networks in Paris, while recurrent units manage the temporal fluctuations of traffic flow. Similarly, the adoption of grid-clustered feature maps by Jin and Noh [71] effectively mitigates the data sparsity problem common in urban environments like Daejeon City. This effectiveness is mainly due to the hierarchical filtering of non-accident "background noise" through convolutional blocks before applying dense layers for risk classification.

In the highway domain, the focus shifts toward mitigating data redundancy and improving interpretability. Huang et al. [86] utilized a SHAP-RFECV feature selection strategy coupled with XGBoost to identify critical precursors from electronic toll collection (ETC) data, noting that speed variations 5–10 min prior to a crash are the most significant predictors. Furthermore, Shang et al. [81] addressed the inherent class imbalance in incident detection through a hybrid SMOTE-RF-RFE-LSTM framework. This limitation in earlier highway models arises from the "curse of dimensionality" and skewed data distributions, which these hybrid selection and oversampling techniques successfully resolve to improve detection rates.

Network-Wide Analysis and Frequency Forecasting

Predicting accident frequency across broader temporal horizons requires models that can internalize complex, non-linear seasonal trends and exogenous influences. Uguz et al. [134], using a dataset from Antalya, demonstrated that CNN-LSTM hybrids consistently outperform linear benchmarks like ARIMA and Prophet, particularly during high-volatility periods such as tourist seasons [134]. This can be attributed to the "synergistic effect" where the CNN captures the intensity of traffic density while the LSTM learns the long-term dependencies of population fluctuations [134].

The integration of behavior-related variables, such as traffic violations, further enhances forecasting accuracy. Wu et al. [135] developed a CNN-GRU fusion model that utilizes city-level violation data to predict at-fault crash driver frequency. Their use of accumulated local effects (ALE) plots revealed that the limitation of purely data-driven models, i.e., the "black box" nature of predictions, can be mitigated by visualizing the non-linear relationship between specific enforcement activities and crash risk [135]. Similarly, the application of LSTM-GBRT by Zhang et al. [136] emphasizes that, while LSTMs are proficient at learning time-series trends, the inclusion of gradient boosted regression trees (GBRT) provides robustness against data fluctuations, making the model more effective for national-level forecasting in China [136].

Methodological Advancements in Accident Severity and Duration

Accident duration and severity prediction have seen a paradigm shift toward multimode data fusion to bridge the gap between numerical traffic metrics and situational

context. Chen et al. [78] demonstrated that a BiGRU-CNN architecture, when integrated with text features extracted via Word2Vec, significantly improves duration prediction on expressways [78]. This is mainly due to the fact that unstructured textual reports contain semantic nuances such as vehicle overturning or hazardous spills that numerical flow data cannot capture [78]. Complementing this, Shang et al. [80] utilized latent Dirichlet allocation (LDA) to transform textual descriptions into topic variables for Bi-LSTM models, identifying that model performance is optimized when textual information is treated as a dynamic updating process during the incident lifecycle [80].

For accident severity, hybrid frameworks emphasize the partitioning of heterogeneous datasets to reveal hidden patterns. Yassin and Pooja [137] employed a hybrid K-means and RF approach, where clustering is used to group similar accident profiles before classification. This limitation in standard classification arises from the high variance within severity datasets, which pre-clustering minimizes, allowing the RF to identify driver experience and light conditions as primary determinants of fatal outcomes [137]. Furthermore, Wu et al. [138] introduced DAMGNet, a multi-scale graph convolutional network with dual attention, to predict accident delay times. This model is particularly effective in the highway domain because it explicitly models how congestion "travels" through a road network, a spatial dependency that grid-based or purely recurrent models often fail to internalize [138].

Across the reviewed studies, a clear consensus emerges: hybrid architectures consistently outperform standalone ML or DL models by decoupling feature extraction from sequence modeling. This methodological synergy allows for the simultaneous processing of spatial road network characteristics and temporal traffic flow dynamics. While urban-focused studies prioritize high-resolution spatiotemporal grids and signalized intersection conflict analysis [139], highway research emphasizes real-time detection through robust feature selection and the integration of ETC data [81]. The future of the field points toward interpretable hybrid frameworks (e.g., SHAP, ALE) and multimodal fusion (text + sensors) to enhance the operational utility of these models for proactive traffic management. The hybrid methods for RTAP described in this section are consolidated in Table 8.

Table 8. Summary of the discussed RTAP hybrid methods.

Ref.	Algorithms	Prediction Task	Data Type	Key Variables	Evaluation Metrics	Strengths	Limitations
[71]	CNN-DNN, accident risk index (ARI)	Accident occurrence and risk severity	Grid-clustered multi-source maps	Driving behaviors (DTGs), mobility density, and facilities	Accuracy, precision, recall, and F1-score	CNN filter mitigates zero-value data bias; accurately identifies hotspots	High computational requirements; weather data not yet integrated into ARI
[78]	Word2Vec-BiGRU-CNN	Traffic accident duration prediction.	Multimode (structured + unstructured text)	Time, location, weather, fatalities, and text descriptions.	Weighted F1-score, precision, and recall	Integration of text data significantly improves DL performance.	High complexity in text feature extraction and simple feature fusion.
[80]	LDA, Bi-LSTM, LSTM	Incident duration with multi-source incomplete data	Natural language text + sensor/environmental data	Weather, POIs, and LDA-processed text topics	RMSE, MAE, and MAPE	Phased sequential model effectively handles missing data	Does not incorporate spatiotemporal correlations of historical crashes.
[81]	RF-RFE, LSTM, BOA, SMOTE	Automatic incident detection (AID)	Numerical data from loop detectors	Volume, speed, and occupancy (actual vs. predicted)	Accuracy, precision, F1, DR, FAR, MTTD, PI	Effective handling of imbalanced data; optimized feature selection	Focuses only on numerical variables; lacks visual or environmental context
[73]	DSHN (CNN, LSTM, ANN)	Short-term spatiotemporal crash prediction	Big traffic data (sensors, weather, holidays)	Flow rate, VKT, WAO, average speed, and holidays	AUC, accuracy, recall, and FAR	Simultaneously captures deep spatial and temporal heterogeneity	Complex architecture; inconsistent impact of weather features
[134]	CNN-LSTM, SARIMA, Prophet	Accident frequency forecasting (tourist season)	Daily time-series accident counts	Historical fatal and injury accident frequencies	RMSE, MAPE, and R-squared	Captures long-term dependency and seasonality for tourism planning	Limited to one specific city; results may not generalize nationally.

Table 8. *Cont.*

Ref.	Algorithms	Prediction Task	Data Type	Key Variables	Evaluation Metrics	Strengths	Limitations
[135]	CNN-GRU	At-fault crash driver frequency	City-level violation and crash records	Speeding, DUI, illegal turns, signal violations	MAPE, AUC, and recall rate	Captures deep spatiotemporal features; uses ALE plots for interpretability	Potential under-reporting of minor crashes; time-lag effects of enforcement
[136]	LSTM-GBRT	Macro-level safety indicator prediction (casualties)	Socioeconomic and traffic macro-data	GDP, highway mileage, population, and number of drivers	RMSLE, R-squared, RMSE, and error rate	Stable and accurate at detecting inflection points in casualty trends	Small annual sample size (19 years); lacks spatial granularity.
[138]	DAMGNet (multi-scale GCN)	Highway accident delay time (congestion)	Real-time traffic and network structure	Accident severity, POIs, road traits, and weather	RMSE and MAE	Models complex spatial dependencies between road segments	Highly dependent on the density and placement of physical sensors
[139]	Bayesian DL (LSTM, Enc-Dec)	Cycle-level traffic conflict frequency	LiDAR point clouds and signal data	Volume, truck ratio, and shock wave characteristics	MSE, MAE, PICP, MPIW, and variance	Quantifies uncertainty; robust to small noise for proactive safety	Sensitivity to certain data noise; validated on a single intersection

The studies summarized in Table 8 represent a methodological shift toward the utilization of hybrid models for accident prediction and management. These approaches demonstrate a superior capability to integrate diverse data sources, including digital sensor data, unstructured text processed via natural language processing (NLP) techniques such as LDA and Word2Vec, and high-resolution LiDAR data [78,80,139]. The functional tasks of these models range from real-time incident detection [81] to predicting accident duration and frequency across various spatiotemporal scales [73,134,135]. Furthermore, they facilitate the identification of safety "hotspots" through innovative indices like the accident risk index (ARI) and uncertainty estimation to ensure proactive safety measures [71,139]. Despite their high efficiency in handling imbalanced datasets and capturing complex heterogeneities [73,81], these models continue to face challenges regarding computational complexity and a heavy reliance on sensor density to ensure robust generalization [71,138].

4.4. Cross-Domain and Cross-Method Analysis

Predicting traffic accidents requires a systematic shift from traditional statistical frameworks to advanced computational approaches that balance predictive accuracy with causal transparency. Studies reveal a recurring pattern whereby DL models excel in extracting complex spatial and temporal features from raw data, whilst ensemble ML models, particularly RF and gradient boosting (XGBoost/LightGBM), remain the most efficient when dealing with scheduled and structured data [113,114,130]. This can be explained by the limitations of statistical models (such as logit and Poisson) in capturing the non-linear relationships and unstable dynamics inherent in smart transport environments, making them less accurate compared to non-parametric frameworks that are not constrained by strict distributional assumptions [87,122,134].

Shared patterns across studies demonstrate that the integration of hybrid models represents the most successful trajectory for enhancing prediction stability. Specifically, combining convolutional neural networks (CNNs) with LSTM networks enables the simultaneous capture of spatial interactions across network links and temporal correlations within traffic flows [23,134]. This superiority stems from the ability of these models to process trajectory data and space–time cubes, moving beyond simple accident frequency counts to a deeper understanding of risky driving behaviors [67,139]. Furthermore, while RF and XGBoost dominate injury severity prediction tasks due to their high efficacy in handling imbalanced datasets [113,114], DL models (such as LSTM, CNN, and GNN) outperform in real-time occurrence prediction within dense urban environments [22,69]. This advantage stems from these frameworks' ability to capture long-term temporal dependencies and

topological interactions between network links, which represents a fundamental limitation in traditional ML models that rely heavily on manual feature engineering [130,139].

Table 9 highlights that ML models demonstrate high effectiveness in predicting incident severity due to their ability to handle non-linear data and provide good interpretability, whilst DL models excel in complex spatiotemporal tasks as a result of their ability to learn directly from raw data. The studies also highlight challenges related to the sensitivity of spatial models to resolution and zero inflation, whilst the growing trend towards hybrid models reflects the importance of integrating textual and structural data to improve prediction accuracy and expand the capabilities of intelligent models.

Table 9. Comparative synthesis of traffic accident prediction approaches across prediction tasks and application domains.

Method	Prediction Task	Domain	Data Type	Main Strength	Main Limitation	Ref.
Statistical Models (Logit, Poisson, ARIMA)	Frequency estimation and injury severity classification	General/rural environments	Structured/tabular data	High interpretability: easy to understand causal relationships and coefficients	Rigid assumptions: struggles with non-linear dynamics and non-normal distributions	[90,94]
Heuristic Methods	Accident severity, frequency forecasting, real-time detection, and monitoring	Urban, highway, and rural road environments	Historical accident, surveillance, GIS, and road geometry data	Enhanced accuracy and computational efficiency under constrained conditions	Regional dependency, computational complexity, and limited driver behavioral data	[104,105]
Ensemble ML (RF, XGBoost, LightGBM)	Severity classification and accident duration	Urban and highway scenarios	Structured data with manual features	Robustness: effective at handling noisy, high-dimensional, and imbalanced data	Feature dependency: relies heavily on manual feature engineering and domain knowledge	[78,114]
Sequential DL (LSTM, GRU, RNN)	Real-time occurrence and risk identification	Highways and traffic flow dynamics	Time-series and high-frequency sensor data	Temporal modeling: superior at capturing long-term and short-term sequential dependencies	Computational cost: requires significant processing power and complex hyperparameter tuning	[22,69]
Spatial/Graph DL (CNN, GNN, ConvLSTM)	Spatial risk hotspots and network-wide safety	Dense urban networks and grid-based areas	Spatial grids, graph structures, and images	Feature extraction: automatically learns complex spatial topologies and local interactions	Black box nature: results are difficult for policymakers to interpret without external tools	[131,132]
Hybrid Models (CNN-LSTM, Multi-Task DNN)	Multi-task prediction (e.g., severity + frequency)	Complex spatiotemporal contexts	Multimodal (text, sensors, structured)	Synergy: combines the strengths of spatial and temporal modeling for higher accuracy	Model complexity: difficult to achieve convergence; requires massive datasets	[80,139]

However, discrepancies in findings emerge when comparing model performance across different application domains. While certain studies demonstrate the superiority of artificial neural networks (ANNs) in classifying accident severity in specific environments [116], other reports indicate that SVM or logistic regression may achieve comparable or superior performance when dealing with small sample sizes or highway domains with less complex variables [25,77]. This inconsistency arises from regional variations in data quality and driving behaviors, as well as selection bias in datasets that may not represent the full spectrum of road conditions. Regarding incident duration prediction, an inconsistency in model ranking is observed; XGBoost outperforms in certain contexts [60], whereas Word2Vec-BiGRU-CNN models excel when integrating unstructured textual data from accident reports [78]. This is attributed to the fact that the incident clearance duration is influenced by logistical and textual factors that traditional sensors fail to capture.

The efficiency of these models is influenced by several critical factors, most notably data quality and the degree of feature engineering. A primary constraint in spatiotemporal modeling arises from the zero-inflation problem, where the majority of spatiotemporal units remain accident-free [79]. This phenomenon can lead to misleading accuracy metrics if balancing techniques, such as SMOTE or focal loss, are not employed [84,117]. Furthermore, predictive accuracy tends to decrease significantly as spatial and temporal resolution increases, suggesting a fundamental trade-off between spatial granularity and predictive power [23]. Additionally, the heavy reliance on data from connected vehicles (CVs) and

advanced sensors limits the generalizability of these models in developing nations, where such infrastructure is currently lacking [130].

Table 10 further indicates that ensemble ML approaches remain highly robust for structured classification tasks such as injury severity prediction, whereas recent studies reveal a growing transition toward hybrid DL architectures for dynamic and high-frequency urban prediction scenarios. The observed variability in algorithmic performance is primarily associated with differences in dataset characteristics, class imbalance, and feature engineering quality across studies. Additionally, the increasing integration of explainable AI techniques, particularly SHAP-based interpretation frameworks, reflects the growing need to balance predictive accuracy with transparency and practical applicability in traffic safety decision-making.

Table 10. Comparative analysis of recurring trends, model variability, and influencing factors across prediction approaches.

Observation	Consistent Findings	Contradictory Findings	Possible Explanation
Algorithm Superiority for Severity Classification	Ensemble tree-based models (RF, XGBoost, LightGBM) consistently outperform traditional logistic regression and standalone SVMs.	Some studies report that ANN or deep MLP architectures achieve higher accuracy (e.g., 90%) than RF on specific regional datasets.	Variation in data volume and balance; ANNs excel in capturing deeper hierarchical features when provided with sufficiently large and balanced training sets [116].
Real-Time Risk and Temporal Modeling	LSTM and GRU networks are superior to static models for capturing temporal dependencies in traffic flow.	Basic RNN often show poor performance or instability compared to LSTM/GRU.	The vanishing gradient problem in standard RNNs hinders their ability to retain long-term memory, which LSTM gates specifically address [73].
Impact of Spatiotemporal Resolution	Integrating spatial features (e.g., POI density, road geometry) through CNNs or GNNs significantly improves network-wide safety prediction.	Predictive performance frequently decreases as the spatiotemporal resolution increases (e.g., moving to 1-h or small grid cells).	Higher resolutions exacerbate the "zero-inflation" problem, where the extreme rarity of accidents in small units makes it difficult for models to distinguish signal from noise [23].
Predicting Accident Duration/Clearance	Hybrid models combining text mining (Word2Vec, BERT) with ensemble learners (RF, XGBoost) achieve the highest accuracy for duration tasks.	Rankings of standard algorithms are inconsistent; some find XGBoost superior while others report SVM or GPR as the best performers on freeway data.	Duration data is typically heavily skewed; kernel-based models like GPR may handle non-linear regression outliers better than discrete tree-based splits in certain contexts [77].
Interpretability vs. Accuracy	DL models generally provide the highest predictive power (AUC up to 0.97) across various tasks.	Policymakers often prefer statistical models or interpretable ML (using SHAP) over highly accurate "black box" DL models.	There is an inherent trade-off; DL models prioritize feature extraction, while safety agencies require causal transparency to design effective interventions [132].

When comparing methodologies, a clear tension emerges between accuracy and explainability. While DL and GNN achieve the highest levels of predictive precision, they often remain "black boxes", making it difficult for stakeholders to decipher the underlying causality of their outputs [133,138]. Conversely, ML models augmented with interpretive tools such as SHAP and ALE provide critical insights into how variables such as vehicle speed or point of interest (POI) density impact accident risk [67,112]. The current shift toward explainable AI (XAI) hybrid models can be explained by the urgent need to translate numerical predictions into tangible safety strategies. Furthermore, RL remains a notable research gap in this field; its application is currently restricted to optimizing post-accident traffic control decisions rather than proactive, predictive modeling.

In conclusion, the synthetic analysis reveals that the future of traffic accident prediction lies in models that integrate the power of DL for automated feature extraction with the

flexibility of ML in handling heterogeneous data. While the primary strengths of current research are concentrated in urban environments supported by multi-source data, significant weaknesses persist in cross-regional generalizability and high computational resource requirements. A persistent research gap remains in the scarcity of studies addressing the long-term impacts of climate change and the transition toward autonomous vehicles within predictive frameworks. Future recommendations point toward enhancing transfer learning models to reduce reliance on big data in regions with limited records, while focusing on the integration of unstructured data to improve the predictive accuracy of accident duration and its secondary effects.

4.4.1. Cross-Study Quantitative Synthesis

A comprehensive quantitative review of the current literature reveals a radical shift in the performance of traffic accident prediction models, with accuracy levels advancing from moderate (in statistical models) to very high (in hybrid and DL frameworks). Quantitative comparisons indicate that model selection is directly proportional to the nature of the predictive task and the spatiotemporal resolution of the data. Specifically, DL and ensemble ML models achieve a statistically significant superiority over traditional statistical frameworks, such as Poisson and logit models, across all performance metrics (accuracy, AUC, and RMSE) [111,130,132]. This can be attributed to the capacity of these computational models to capture complex, non-linear relationships that linear models fail to accommodate, particularly when processing massive, multi-source datasets [84,87].

Performance Analysis by Forecasting Task

In accident severity prediction tasks, RF and gradient boosting models (XGBoost/LightGBM) emerge as benchmark tools, with classification accuracies ranging between 75% and 85% across most studies [114,120]. However, findings indicate that ANN can achieve accuracies as high as 90% when processing massive, balanced datasets [116]. This superiority is attributed to the ability of ensemble learners to minimize variance and bias, making them more robust in classifying minority classes (such as fatal accidents), particularly when integrated with data-balancing techniques like SMOTE [84,117].

Regarding real-time occurrence prediction, DL models, specifically LSTM and CNN, have recorded the highest AUC values, reaching up to 0.976 in certain studies [67]. This exceptional performance stems from the ability of these frameworks to automatically extract features from trajectories and spatiotemporal relationships, eliminating the need for extensive manual feature engineering [65,73]. Conversely, it is observed that the performance of these models decreases significantly as spatiotemporal resolution increases. This decline is attributed to the exacerbation of the zero-inflation problem within smaller units, where the rarity of events makes accurate prediction increasingly challenging [23].

Comparative Performance and Spatiotemporal Complexity

A comparison of the studies shows that spatial ML models achieve improvements in error metrics (MAE/RMSE) of between 12% and 45% compared to global models [87,111]. This can be explained by the ability of spatial models (such as GW-RF and GSVR) to address spatial heterogeneity and account for the influence of local factors (such as intersection density) that vary from neighborhood to neighborhood [107,111]. In incident duration prediction tasks, hybrid models that integrate natural language processing (NLP) with ML demonstrated clear superiority, with models such as Word2Vec-BiGRU-CNN performing best when combining textual reports with structural data [78,80] (Table 11).

Table 11. Comparative synthesis of prediction performance across methods and tasks.

Prediction Task	Method Category	Key Metrics (Averages)	Main Finding
Severity Prediction	Ensemble ML (RF, XGB)	Accuracy: 76–84%, F1: 0.78	Robust for structured data; handles imbalance well [86,120].
	Statistical (Logit/Probit)	Accuracy: 60–70%	High interpretability but lower predictive power [25,132].
Occurrence (Real-Time)	DL (LSTM, CNN)	AUC: 0.85–0.97	Superior in capturing temporal trajectories [22,67].
	Heuristic Optimal Scheduling	Efficiency Gain: 17–40%	Maximizes detection efficiency in large-scale surveillance by dynamically reallocating resources to high-risk nodes [103].
	Hybrid Models	AUC: 0.91–0.98	Synergistic effect of spatial and temporal features [73,132].
Frequency Estimation	Spatial ML (GW-RF, GSVR)	MAE/RMSE: 15–30% Lower	Effectively models spatial non-stationarity [87,110].
Accident Duration	Hybrid (NLP + DL/ML)	F1: 0.52, MAE: 13–16 min	Unstructured text data improves situational awareness [78,80].

Influencing Factors and Technical Trade-Offs

Quantitative comparison reveals a fundamental trade-off between "accuracy" and "interpretability". Whilst DL models achieve high complexity and superior accuracy, they lack the causal transparency required for sovereign decision-making in road safety [133,138]. To bridge this gap, recent studies have turned to interpretability tools such as SHAP, which have demonstrated that variables such as "vehicle speed" and "traffic flow density" are the most influential in high-performance models [112]. Furthermore, model complexity and generalizability are linked to data quality; studies show that models trained on smart city data (such as drones and connected vehicles) achieve significantly higher accuracy than those relying solely on historical accident data [67,135] (Table 12).

Table 12. Average performance metrics and key influencing factors across studies.

Factor	Impact on Performance	Causality/Explanation
Data Imbalance	Decreases recall (often < 0.40)	Standard loss functions favor the majority class [60,84]
Feature Engineering	Increases ML accuracy by 10–20%	Manual feature selection reduces noise in small datasets
Spatiotemporal Res.	Higher resolution = lower accuracy	Due to zero-inflation and extreme rarity of events [23]
Model Complexity	Higher complexity = higher accuracy	DL models learn hierarchical features automatically [130]
Interpretability (SHAP)	No impact on accuracy	Crucial for identifying causal risk factors for policy [112]

In conclusion, this cross-sectional quantitative analysis shows that hybrid models are the most effective at striking a balance between spatial and temporal feature extraction. However, a persistent challenge remains in the form of heavy reliance on data quality and the need to balance unequal classes to ensure prediction reliability. The current gap lies in the scarcity of models that achieve "cross-domain generalization", as most models remain confined to the geographical scope on which they were trained, calling for a shift towards "transfer learning" in future research.

4.5. Common Evaluation Indicators

The evaluation of RTAP models relies on a variety of metrics tailored to the specific nature of the prediction task, whether it involves discrete classification (such as severity levels) or continuous regression (such as incident duration or frequency).

4.5.1. Classification Metrics for Severity and Occurrence

For tasks involving the classification of accident severity (e.g., slight, serious, fatal) or predicting whether an accident will occur, accuracy, precision, and recall (sensitivity) are the most fundamental metrics employed across various studies [65,74,77,81]. However, because traffic accident data is often highly imbalanced, with non-accident cases or minor

injuries far outnumbering fatal ones, standard accuracy can be a misleading indicator of performance [81,117]. To address this, the F1-score, which is the harmonic mean of precision and recall, is widely used to ensure a balanced evaluation of the model's effectiveness in identifying rare but critical crash events [68,71,74,78,81]. Studies like [74] utilize a specific variant called *Micro_F1score* for comprehensive multi-class evaluation. Additionally, the area under the curve (AUC) of the receiver operating characteristic (ROC) is frequently cited to measure the model's overall ability to distinguish between different severity classes across various thresholds [25,65,67,81,132].

Accuracy and the Confusion Matrix: Overall accuracy is the most common metric used to evaluate statistical and ML models. The confusion matrix is also used as a key tool for identifying correct and incorrect predictions for each class individually.

$$\text{Accuracy} = \frac{TN + TP}{TN + TP + FP + FN} \tag{1}$$

where TP and TN represent correct hits for positive and negative classes, and FP and FN denote the respective prediction errors.

Precision and Recall (Sensitivity): Recall measures the model's ability to detect all actual incidents and is a critical indicator in road safety. Study [113] noted the use of recall as a key success metric in injury severity classification.

$$\text{Precision} = \frac{TP}{FP + TP} \tag{2}$$

$$\text{Recall} = \frac{TP}{FN + TP} \tag{3}$$

where TP and TN represent correct hits for positive and negative classes, and FP and FN denote the respective prediction errors.

F1-score: Because accident datasets often exhibit class imbalance, the F1-score is employed as a harmonic mean to balance precision and recall. In study [74], an enhanced version known as the micro-F1-score was utilized to provide a comprehensive evaluation across multiple severity classes.

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

Area Under the Curve (AUC-ROC): This metric is utilized to evaluate model performance across various decision thresholds. A value approaching 1 indicates a superior capability to distinguish between accident and non-accident instances.

$$\text{AUC-ROC} = \int_0^1 \text{TPR}(\text{FPR})\, d(\text{FPR}) \tag{5}$$

where

- TPR (true positive rate or sensitivity) is defined as

$$\text{TPR} = \frac{TP}{TP + FN} \tag{6}$$

- FPR (false positive rate) is defined as

$$\text{FPR} = \frac{FP}{FP + TN} = 1 - \text{Specificity} \tag{7}$$

4.5.2. Regression Metrics for Duration and Risk Frequency

When models are designed to predict continuous variables, such as the time required to clear an incident or the expected frequency of crashes on a road segment, error-based metrics predominate. The mean absolute error (MAE) and root mean square error (RMSE) are the primary indicators used to quantify the magnitude of prediction errors [23,80,87,130,134,138] (Table 13). For example, the authors of [138] employ these to evaluate highway accident delay times, while in [23], they are used alongside the mean absolute percentage error (MAPE) to assess citywide short-term crash risks [23]. The MAPE is particularly valued because it provides a percentage-based view of error, making it easier to compare model performance across different datasets [81,130]. To measure how well a model captures the underlying variability of the data, the coefficient of determination (R^2) is often reported, with values closer to 1 indicating superior fit [74,78,134].

Table 13. Theoretical equations of evaluation indices used for regression where N is the number of observations, y_i is the actual (observed) value, f_i is the predicted value, and $\bar{y}$ is the mean of the observed values.

Evaluation Parameter	Equation	Characteristics
Mean Absolute Error (MAE) and Root Mean Square Error (RMSE)	$\mathrm{MAE}(y, f_i) = \frac{1}{N}\sum_{i=1}^{N}\lvert y_i - f_i\rvert$ $\mathrm{RMSE}(y, f_i) = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - f_i)^2}$	These serve as fundamental benchmarks for evaluating duration and frequency prediction models. Study [138] further incorporated metrics such as the mean absolute error (MAE) and root mean square error (RMSE) to enhance the accuracy.
MAPE, or Mean Absolute Percentage Error	$\mathrm{MAPE}(y, \hat{y}) = \frac{100\%}{N}\sum_{i=0}^{N-1}\frac{y_i - f_i}{y_i}.$	This indicator facilitates the comparison of model performance across datasets of varying scales by expressing the error as a percentage.
Coefficient of Determination (R^2)	$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - f_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2}.$	This metric is utilized to quantify the proportion of variance in the target data that is predictable from the model. It is a widely adopted measure for evaluating the performance of duration and accident frequency forecasting models.

4.5.3. Specialized and Real-Time Performance Indicators

In real-time incident detection and active traffic management, specialized metrics are required to evaluate operational efficiency (Table 14). The false alarm rate (FAR) or false positive rate (FPR) is a critical metric for ensuring that detection systems do not overwhelm operators with incorrect alerts [22,81]. In [81], the mean time to detect (MTTD) is introduced, which measures the delay between the occurrence of an incident and its automated detection, as well as a composite performance index (PI) that mathematically combines the detection rate, FAR, and MTTD to provide a single success value for the system [81]. Furthermore, the authors of [23] utilized the prediction accuracy rate (PAR) to evaluate the precision of identifying high-risk "hotspots" in spatiotemporal risk maps [81]. From an economic perspective, the authors of [25] proposed cost-based metrics such as the overall prediction error (OPE) and specific prediction error (SPE), which weigh the accuracy of predictions by the estimated monetary loss associated with different crash severities [25].

Table 14. Theoretical equations of evaluation indices used for real-time incident detection.

Evaluation Parameter	Equation	Characteristics
False Alarm Rate (FAR)	$\mathrm{FAR} = \frac{FP}{FP + TN}$	A vital indicator in real-time detection systems used to ensure that control rooms are not overwhelmed by erroneous alerts.
Mean Time to Detection (MTTD)	$\mathrm{MTTD} = \frac{1}{N}\sum_{i=1}^{N}(t_i^{\mathrm{det}} - t_i^{\mathrm{true}})$ where N is the number of events, t_i^{true} is the actual time of occurrence of event i, t_i^{det} is the time at which the event is detected.	This metric quantifies the duration between incident occurrence and automated detection, representing a pivotal benchmark for performance in research [81].
Prediction Accuracy Rate (PAR)	$\mathrm{PAR} = \frac{TP + TN}{TP + TN + FP + FN}$	This metric is utilized in network-level risk prediction models, such as STCL-Net, to evaluate the accuracy of identifying high-risk areas (hotspots).
Performance Index (PI)	$\mathrm{PI} = \sum_{i=1}^{N} w_i\, A_i$ where A_i is the number of accidents of type i, w_i is the weight associated with accident severity (e.g., fatal, injury, minor), N is the number of accident categories.	Some studies have developed composite indices that integrate the detection rate (DR), false alarm rate (FAR), and mean time to detection (MTTD) into a single mathematical formulation to evaluate the system's overall performance.

4.5.4. Statistical Model Selection Criteria

For traditional statistical models, such as Poisson or negative binomial regressions, selection is often guided by information criteria that penalize model complexity. The Akaike information criterion (AIC) and Bayesian information criterion (BIC) are standard tools for comparing model fits, where lower values indicate a better balance between accuracy and parsimony [89,130]. Additionally, McFadden's R^2 is frequently used in logit-based severity models to assess the relative improvement of the proposed model over a null model [89].

5. Discussion and Research Gaps

The synthesized results of this systematic review underscore a paradigm shift in traffic safety analytics, where the integration of high-fidelity data and advanced computational frameworks has transitioned from reactive modeling to proactive risk identification. This discussion interprets the multifaceted relationships between model architectures, data sources, and operational requirements within the ITS ecosystem.

5.1. Comparative Performance of Prediction Models

Analytical Comparison of Predictive Methodologies: The analytical comparison reveals a significant shift away from traditional statistical models (e.g., Poisson and logit) toward modern computational approaches. This transition is driven by the fact that traditional models rely on rigid distributional assumptions and linear prescriptions that fail to capture the "non-linearity" and high-dimensional complexity inherent in smart city traffic flows [87,91,130]. While these models offer high "interpretability" and enable direct causal analysis of statistical significance, their inability to handle heterogeneous data has rendered them more "descriptive" than "predictive" in complex environments [91]. Nevertheless, advanced statistical frameworks like SARIMA remain the most robust tools for analyzing long-term seasonal trends in safety data [91,130].

Regarding heuristic methods and optimization algorithms (e.g., GWO and GA), their continued relevance reflects the necessity of addressing structural complexity and multi-objective constraints within dynamic traffic environments [104]. Their increasing success is attributed to their superior ability to circumvent local optima and automate the hyper-parameter tuning of traditional models like SVM and neural networks. This enhances computational efficiency under resource-constrained conditions and facilitates real-time adaptive decision-making [81,106].

In contrast, ensemble ML models, such as RF, XGBoost, and LightGBM, have emerged as the gold standard for severity classification and accident frequency tasks on structured tabular data. Their effectiveness stems from their ability to manage noisy data and reduce variance through aggregation mechanisms, achieving classification accuracies up to 85% higher than statistical counterparts [114,116,117].

The most prominent methodological shift is evidenced by the superiority of DL frameworks—specifically LSTM, GRU, and GNN—in complex spatiotemporal tasks. These architectures excel at modeling the "transitional state" preceding a collision by capturing instantaneous fluctuations in speed and density that static historical data often overlook [22,67,69]. Notably, graph neural networks (GNNs) represent the state of the art in network safety analysis by modeling synchronous correlations between adjacent traffic links, yielding substantial performance gains [131,138].

Furthermore, the growing adoption of RL signifies a fundamental shift from static modeling to goal-oriented dynamic frameworks. RL outperforms deep models in interactive scenarios by treating environmental interactions as a Markov decision process (MDP), enabling proactive risk assessment and the adaptation of safety policies (such as variable speed limit—VSL) in response to real-time feedback [26,128]. However, the "reality gap" and the prohibitive computational cost of training remain significant barriers to large-scale field deployment [26,127].

Finally, hybrid models (e.g., CNN-LSTM or STCL-Net) represent the pinnacle of methodological synergy. This trend is driven by the need to address "unobserved heterogeneity". These models allow for the integration of automated spatial feature extraction with temporal modeling, alongside the incorporation of natural language processing (NLP) to extract qualitative context from accident reports. Such integration has increased the duration prediction accuracy by up to 24%, effectively closing the gaps left by standalone models [4,73,80,134].

Given the complexity of these predictive frameworks, verifying their efficiency requires a robust assessment strategy. Consequently, the current literature employs a diverse matrix of evaluation metrics to ensure the robustness and accuracy of research findings. In tasks involving accident occurrence and severity classification, the accuracy, recall, and area under the curve (AUC) are the predominant metrics [77,81]. However, the F1-score has emerged as a critical benchmark for addressing the "class imbalance" problem, ensuring the effective detection of rare and high-risk events that overall accuracy metrics might overlook [81,132]. For continuous prediction tasks, such as accident duration and frequency, error metrics including the MAE, RMSE, and MAPE serve as the standard criteria for performance evaluation. Concurrently, the coefficient of determination (R^2) and information criteria (AIC/BIC) are utilized to methodologically differentiate between the explanatory power of statistical models and the predictive prowess of computational approaches [91,130,134]. Table 15 below provides a synthesis of the main categories of data-driven forecasting methods reviewed, supporting a critical comparison of their strengths and limitations.

Table 15. Comparison of forecasting method categories with selected algorithms.

Criteria	Statistical Models	Heuristic Methods	Machine Learning (ML)	RL Models	Deep Learning (DL)	Hybrid Models
Interpretability	**Very High**. Directly identifies variable significance via coefficients.	**Moderate to High**. Models like CPO-RF and PSO-BPNN use factor analysis or index importance to explain variable impacts.	**Moderate.** Often requires external tools like SHAP to explain "black box" logic.	**Low.** Operates as a "black box," making decision logic difficult for policymakers to trace.	**Low.** Operates as a complex "black box", making decision logic difficult to trace.	**Very Low.** Extreme architectural complexity further obscures transparency.
Non-Linearity	**Poor**. Generally assumes linear relationships between variables.	**Excellent.** Effectively captures complex, non-linear relationships in crash data through kernels or tree structures.	**Very Good.** Capable of capturing high-dimensional non-linear patterns.	**Excellent.** Designed specifically to map complex, non-linear traffic dynamics in real time.	**Excellent.** Specifically designed to map complex, non-linear traffic dynamics.	**Superior.** Optimizes the capture of multifaceted non-linear interactions.
Spatiotemporal Dependency	**Low.** Often struggles with spatial heterogeneity and temporal lags.	**High.** Integrates variables like time of day, season, and GIS-based location data.	**Moderate.** Can incorporate spatial features but often requires manual engineering.	Automatically extracts dependencies through real-time state perception (e.g., DQN, PPO).	**Very High.** Automatically extracts spatial (CNN) and temporal (LSTM) dependencies.	**Superior.** Effectively integrates multi-scale spatial and long-term temporal features.
Data Requirements	**Small to Moderate.** Efficient for smaller, univariate datasets.	**Extensive.** Requires detailed environmental, roadway, and socioeconomic data (e.g., GDP, population density).	**Moderate to Large.** Performance scales with feature richness and sample size.	**Very Large.** Requires massive amounts of real-time traffic and environmental data to avoid overfitting.	**Very Large.** Requires massive datasets to ensure model stability and avoid overfitting.	**Very Large.** Demands extensive multi-source data (textual, numerical, and visual).
Computational Resources	**Low.** Can be processed quickly on standard CPUs.	**Significant.** Meta-heuristics require iterative training, often necessitating robust hardware for big data.	**Moderate.** Training times vary; ensemble methods may require more memory.	**High.** Necessitates high-performance GPUs for intensive simulation training (SUMO, CARLA, Unity3D).	**High.** Necessitates high-performance GPUs for training deep architectures.	**Very High.** Significant computational overhead for multi-stage optimization and fusion.
Common Models	Logistic regression, Poisson, SARIMA, ES.	ModelsCPO-Random Forest, SSA-SVM, PSO-BPNN.	RF, SVM, XGBoost, LightGBM.	DQN, D3QN, PPO, SAC, DDPG, Q-Learning.	LSTM, CNN, GCN, ConvLSTM, GRU.	CNN-LSTM, STCL-Net, BOA-LSTM, TF-IDF-RF.
Strengths	Simple, transparent, and ideal for establishing statistical significance.	Avoids local optima and automates hyperparameter selection for high accuracy.	Excellent balance between predictive accuracy and computational efficiency.	Automatic feature extraction and superior accuracy (95.2% in R1) in dynamic environments.	Automatic feature extraction and superior accuracy in short-term forecasting.	Combines the strengths of multiple architectures and data types.
Limitations	Predefined functional forms; poor transferability to new regions.	Often lacks driver behavioral indicators and requires regional recalibration.	Requires intensive manual feature engineering; sensitivity to data noise.	Computationally demanding; reliance on high-fidelity simulations over real-world data.	Computationally demanding; results are difficult for policymakers to interpret.	Highly complex implementation; increased risk of processing latency.

5.2. Impact of Big Data, IoT, and Intelligent Transportation Systems

Data quality and infrastructure represent the cornerstone of success for traffic accident prediction systems within smart city environments. Research has evolved from an exclusive reliance on manual historical government records toward the integration of multi-source big data [71]. Evidence suggests that utilizing microscopic sources such as connected vehicle (CV) data, and digital tachograph (DTG) logs enables the monitoring of risky driving behaviors, such as harsh braking and sudden acceleration, mere seconds before a collision occurs. This provides a predictive accuracy that traditional stationary sensors cannot achieve. This shift is driven by the transition from static analysis to "proactive prevention", which relies on understanding real-time traffic flow dynamics [71,133].

The preprocessing stage represents the most critical technical challenge in ensuring model reliability, as accident datasets suffer from structural "class imbalance" accidents being rare events compared to normal traffic flow [84,109]. Findings indicate that employing resampling techniques, such as SMOTE or negative sampling, is indispensable to prevent model bias toward the majority class and to ensure the recall of critical and fatal cases. Furthermore, it is essential to clean and calibrate data from Internet of Things (IoT) sensors, which are prone to malfunctions or missing values. This necessitates resilient

frameworks to manage "heterogeneity" and ensure the stability of deep learning model performance [116,120].

The practical deployment of these systems faces significant challenges related to latency and data availability, as deep learning models often require substantial computational resources that may hinder the real-time responsiveness required in ITSs. In this context, edge AI has emerged as a technical solution to reduce processing delays by performing computations closer to the data source [65], thereby enabling the transmission of immediate warnings through vehicle-to-everything (V2X) communications without delays that could compromise traffic management decisions [84,133]. Furthermore, the integration of unstructured data through natural language processing (NLP) techniques, such as Word2Vec and latent Dirichlet allocation (LDA), contributes to decoding contextual information embedded in accident reports that cannot be captured by purely numerical data. This integration enhances the accuracy of road clearance time prediction and improves overall traffic incident management efficiency [4,80].

Scalability remains closely associated with the level of spatial and temporal resolution adopted, as several studies have attempted to partition transportation networks into spatiotemporal grids to simplify complex traffic relationships [23,71]. However, increasing the granularity of these networks often exacerbates the "zero inflation" problem, thereby introducing additional challenges to the reliability and robustness of predictions in large-scale urban environments. Consequently, the success of the next generation of ITSs depends on establishing a sustainable balance between real-time data accuracy, model transparency through XAI, and the protection of drivers' data privacy to ensure safe and effective real-world deployment [71].

5.3. Bibliometric Insights and Emerging Trends

The bibliometric analysis of research on intelligent transportation and road safety reveals a significant paradigm shift from traditional statistical modeling to high-dimensional AI. ML and DL have emerged as the central pillars of the field, acting as motor themes with high technical density and strategic importance. The keyword co-occurrence network identifies a strong nexus between these AI tools and practical applications such as risk assessment, traffic management, and crash prediction. Furthermore, the prevalence of terms like "human forecasting" and "highway accidents" in study titles indicates a specialized focus on integrating human behavioral data with predictive analytics to enhance safety protocols. While ITSs remain a foundational basic theme, the emergence of specialized niche themes like autonomous vehicles, LSTM, and GRU suggests that future research is gravitating towards real-time, automated decision-making and deep sequence modeling to handle complex spatiotemporal traffic data.

5.4. Challenges and Limitations

The class imbalance problem emerges as a fundamental structural barrier in predictive modeling, as traffic accidents are inherently "rare events" compared to normal traffic flow [71,131]. This constraint arises from the tendency of standard models to bias toward "non-accident" predictions, rendering high accuracy rates deceptive unless augmented by resampling techniques such as SMOTE or specialized loss functions like focal loss to ensure the recall of severe and fatal crashes [81,117]. Furthermore, zero-inflation becomes prominent when increasing spatiotemporal resolution (e.g., grid cells under 200 m). In such cases, the lack of accident data in most units destabilizes deep learning models and complicates the separation of signal from noise [73].

Technically, latency and computational overhead represent a critical gap in current ITS frameworks. While DL models achieve superior accuracy, their structural complexity

demands substantial computational resources, which may impede the real-time response necessary to provide proactive warnings within narrow 30-s windows [65,84]. Additionally, the vulnerability of sensor reliability remains a concern, as models often face predictive collapse during data loss or sensor failure, necessitating the development of robust frameworks for managing missing and heterogeneous data [22].

Methodologically, limited transferability remains one of the greatest obstacles to global model generalization. Models trained in specific geographic settings often lose precision when deployed in different traffic environments due to variances in road geometry, regional driving behaviors, and climatic conditions [22,138]. This disparity is largely driven by the dominance of the human element, responsible for 80% to 90% of accidents, which renders "static" models incapable of accounting for unpredictable behavioral fluctuations [118].

Finally, the "black box" nature of deep models hinders their adoption in policymaking. Despite achieving high-performance metrics (AUC > 0.90), these models lack the causal transparency required by urban planners to justify large-scale safety investments [117,132]. Balancing accuracy with interpretability requires the integration of advanced tools like SHAP to transform numerical outputs into actionable insights, a trend that still faces technical hurdles in real-time integration, alongside ethical and legal constraints regarding data privacy [65].

6. Conclusions

The field of RTAP has undergone rapid evolution, transitioning from traditional statistical models (e.g., logit and NB) focused on interpreting linear relationships to more sophisticated ML and DL algorithms. This body of literature addresses diverse tasks, including the prediction of crash frequency, injury severity, and accident duration. Studies have increasingly relied on hybrid data sources, integrating structured data (e.g., road and weather attributes) with unstructured data (e.g., textual crash reports and CCTV imagery) to simulate the complexity of real-world traffic environments. **Findings:** The results indicate that DL and ensemble models (e.g., TASP-CNN, STCL-Net, and XGBoost) achieve superior predictive accuracy compared to statistical benchmarks, particularly when handling complex spatiotemporal dependencies and high-dimensional data. Factors such as roadway characteristics, lighting conditions, and hazardous driving behaviors (e.g., sudden braking) have been identified as the most significant predictors of crash occurrence and severity. Furthermore, the integration of unstructured textual data has demonstrated a remarkable capacity to enhance the accuracy of accident duration prediction and provide contextual details absent from numerical datasets. **Challenges:** The literature identifies fundamental challenges hindering predictive efficiency, most notably the issue of class imbalance, which remains a primary obstacle; accidents are inherently rare events compared to normal traffic flow volumes. This disparity often leads to a model bias toward the majority class, resulting in a failure to accurately detect infrequent collision events. Consequently, researchers have heavily relied on sampling techniques such as SMOTE to mitigate this issue. Furthermore, the complex nature of DL models, often described as "black boxes", poses a significant barrier to institutional trust. Policymakers require a clear understanding of risk factors and the underlying logic behind predictions to adopt effective preventive measures, a transparency that traditional deep architectures lack. Additionally, transferability is a persistent concern; studies indicate that models trained on specific road networks or cities lose significant accuracy when deployed in different geographical contexts due to unobserved data heterogeneity. Finally, the technical demands of latency and the substantial computational resources (GPUs) required to operate hybrid architectures remain a constraint, impeding the broad real-time implementation of these models within field-based traffic control systems. **Future Directions:** Future research

should focus on integrating XAI techniques, such as SHAP and LRP, to enhance model transparency. Priority should also be given to leveraging connected vehicle (CV) data to provide broader, real-time risk coverage. It is essential to develop multi-task learning frameworks capable of predicting multiple severity types simultaneously and to move toward dynamic modeling that updates predictions in real time as new information emerges from the crash site. Finally, there is an urgent need for models with high transferability across diverse geographic environments to reduce the necessity of retraining from scratch.

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Abbreviations

The following abbreviations are used in this manuscript:

RTAP	Road Traffic Accident Prediction
ML	Machine Learning
DL	Deep Learning
NBR	Negative Binomial Regression
GPR	Generalized Poisson Regression
RL	Reinforcement Learning
ANN	Artificial Neural Network
CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
LSTM	Long Short-Term Memory
GRU	Gated Recurrent Unit
SVM	Support Vector Machine
RF	Random Forest
LR	Logistic Regression
DT	Decision Tree
KNN	K-Nearest Neighbor
GBDT	Gradient Boosting Decision Tree
ConvLSTM	Convolutional Long Short-Term Memory
BiLSTM	Bidirectional Long Short-Term Memory
XGBoost	Extreme Gradient Boosting
AI	Artificial Intelligence

ARMA	Autoregressive Moving Average
ARIMA	Autoregressive Integrated Moving Average
BRR	Bayesian Ridge Regression
DBN	Deep Belief Network
GB	Gradient Boosting
LGBM	Light Gradient Boosting Machine
LR	Linear Regression
NB	Negative Binomial
BRF	Balanced Random Forest
GW-RF	Geographically Weighted Random Forest
MLP	Multi-Layer Perceptron
MLR	Multiple Linear Regression
SZINB	Spatial Zero-Inflated Negative Binomial
RMSE	Root Mean Square Error
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
AUC	Area Under the Curve
ROC	Receiver Operating Characteristic
F1-score	Harmonic Mean of Precision and Recall
Accuracy	Prediction Accuracy
Precision	Positive Predictive Value
Recall	Sensitivity
FAR	False Alarm Rate
MTTD	Mean Time to Detection
AUC-ROC	ROC Curve and Area Under the Curve
FPR	False Positive Rate
TP/FP/TN/FN	True Positive, False Positive, True Negative, and False Negative
R^2	Coefficient of Determination
ITS	Intelligent Transportation System
GPS	Global Positioning System
IoT	Internet of Things
API	Application Programming Interface
SUMO	Simulation of Urban MObility
FM2GI	Feature Matrix to Gray Image algorithm
SMOTE	Synthetic Minority Oversampling Technique
ARI	Accident Risk Index
CV	Connected Vehicle
AFT	Accelerated Failure Time

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